

Vision for Pollution: Deep Learning to Detect Visual Pollution

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Abstract

Urban visual pollution, arising from sources such as billboards, graffiti, garbage, construction materials, potholes, and overhead wires, poses significant challenges to city aesthetics, urban identity, and public well-being. Traditional assessment methods rely on subjective surveys and manual scoring, which limit scalability and measurable accuracy. Recent developments in deep learning have enabled automated detection, mostly using object detection models that produce only coarse estimates of pollutant coverage. This dissertation examines semantic segmentation as a more detailed and interpretable solution for visual pollution detection and quantification.

A custom pixel-level annotated dataset was developed, covering six major pollutant types, which enabled thorough evaluation of both conventional supervised models (U-Net, DeepLabV3+, FPN, and SegFormer) and open-vocabulary vision-language models (CLIPSeg and OVSeg). Comparative experiments showed that supervised segmentation models work well when pollutant boundaries are clear, but their ability to generalize to new classes or domains is limited. Open-vocabulary segmentation offers greater flexibility by allowing prompt-based identification of pollutants, although its success depends on careful prompt selection and probability thresholding, and sometimes results in less precise boundaries. Further fine-tuning may also be needed to improve performance.

To provide region-level pollution scores, the Visual Pollution Index (VPI) framework was adapted to utilize segmentation masks, enabling quantification of pollutant severity, diversity, and frequency in urban environments. Case studies using Google Street View images illustrate how segmentation-based scoring captures differences in visual clutter between cities.

Despite these advances, challenges remain regarding dataset diversity, robustness under realistic conditions, and the absence of ground truth for large-scale urban images. Future work should focus on expanding annotated datasets and strengthening model generalization and robustness. Overall, this study demonstrates that semantic segmentation offers a promising foundation for flexible and detailed assessment of urban visual pollution.

Research Ethics Approval

This project was planned in accordance with the Informatics Research Ethics policy. It did not involve any aspects that required approval from the Informatics Research Ethics committee.

Declaration

I declare that this thesis was composed by myself, that the work contained herein is my own except where explicitly stated otherwise in the text, and that this work has not been submitted for any other degree or professional qualification except as specified.

(Yoseph Widistika Rangga Prakoso)

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Table of Contents

1	Introduction	1
1.1	Motivation	1
1.2	Research Questions	3
1.3	Contributions	4
1.4	Dissertation Structure	4
2	Related Works	6
2.1	Traditional Visual Pollution Assessment	6
2.2	Object Detection Models for Visual Pollution	7
2.3	Semantic Segmentation Models	8
2.4	Open-Vocabulary Segmentation Models	9
3	Data	10
3.1	Dataset Collection and Sources	10
3.2	Class Distribution	12
3.3	Pollutant Types per Image	13
3.4	Class Co-occurrence	13
3.5	Pollutant Coverage Analysis	14
3.6	Image Brightness Analysis	15
3.7	Model-assisted Annotation and Mask Refinement	16
4	Methodology	17
4.1	Conventional Supervised Segmentation Models	17
4.1.1	U-Net	17
4.1.2	DeepLabV3+	18
4.1.3	Feature Pyramid Network (FPN)	18
4.1.4	SegFormer	19
4.2	Open-Vocabulary Segmentation Models	19

4.2.1	CLIPSeg	19
4.2.2	OVSeg	20
4.2.3	Prompt Engineering for Binary and Multi-Class Segmentation	20
4.3	Data Preprocessing and Augmentations	22
4.4	Optuna Hyperparameter Search	22
4.5	Experimental Setup	23
4.6	Loss Function and Evaluation Metric	24
4.7	Robustness Testing	24
5	Results and Analysis	25
5.1	Conventional Supervised Models	25
5.1.1	Final Hyperparameters from Optuna	25
5.1.2	Model Comparison	26
5.1.3	Robustness Testing	28
5.2	Open-Vocabulary Models	29
5.2.1	CLIPSeg	30
5.2.2	OVSeg	32
5.3	Discussion	33
5.3.1	Supervised vs. Open-Vocabulary Models	33
5.3.2	The Role of Prompt Engineering	34
5.3.3	Robustness and Deployment Considerations	35
6	Visual Pollution Scoring	36
6.1	Scoring Framework Adaptation	36
6.1.1	Severity-Weighted Score (SWS)	37
6.1.2	Counting Categories Ratio (CCR)	37
6.1.3	Pollution Ratio (PR)	37
6.2	Example Application	38
6.2.1	Data Collection	38
6.2.2	Results and Discussion	38
7	Conclusion	40
	Bibliography	41
A	Supplementary Information	46
A.1	Model Architectures	46

A.2	SegFormer	50
A.3	CLIPSeg	53
A.4	OVSeg	54
A.5	Visual Pollution Scoring	55

Chapter 1

Introduction

1.1 Motivation

In recent years, while efforts to build more modernized cities have brought significant economic and social benefits, the negative footprints on the environment are becoming ever-present. This includes various forms of pollution, defined as the accumulation of contaminants or other harmful substances that adversely affect the environment and human health [15]. Traditional forms of environmental pollution, such as air, water, and noise pollution, have long been considered as one of the most dangerous threats to human life quality and studied due to their well-established impacts on human health and ecosystem stability [18, 23, 29]. However, as the world continues to evolve, another form of environmental degradation has emerged: visual pollution.

Visual pollution refers to the deterioration or disruption of natural and man-made landscapes that ruins the aesthetic appeal of an area [8]. Some examples of visual pollutants include billboards, tangles of electric wires, street litters, construction materials, graffiti, cellphone towers, worn out buildings, and other urban clutter [20]. Unlike conventional pollutants, visual pollution does not directly contaminate natural resources or cause acute health effects. However, these pollutants reduce aesthetic quality, impair visual enjoyment of spaces, and disrupt urban identity and livability [15]. Another concern is that vulnerable populations, such as lower-income groups, the elderly, and children, are disproportionately affected by visual pollutants, facing heightened stress, anxiety, and diminished quality of life from high-clutter environments [19].

Visual pollution has been linked to a range of detrimental effects on human well-being, particularly in cognitive and physiological domains. Excessive visual clutter, such as tangled wires, oversized signage, and dense roadside advertising, not only

detracts from environmental aesthetics but also imposes mental and sensory strain on individuals [40]. Prolonged exposure to such overstimulating environments can lead to eye fatigue, reduced concentration, and directed-attention fatigue, a condition where the brain's ability to maintain focus is impaired [44]. Beyond cognitive impacts, visual pollution also poses safety risks by impairing driving performance. Studies show that drivers in visually cluttered environments tend to make more lateral glances, to the left and right sides of the road, compared to those in less cluttered settings [34]. In particular, the presence of billboards in high-clutter areas significantly increases the frequency and duration of driver glances away from the road. This supports earlier findings suggesting that exposure to irrelevant visual stimuli can overwhelm drivers' cognitive processing, resulting in information overload and reduced ability to detect road hazards [25].

In light of these growing impacts, visual pollution is no longer merely an aesthetic concern but a pressing urban governance issue that demands immediate intervention. In many cities, it is exacerbated by weak regulatory enforcement, poor urban planning, and the unchecked proliferation of outdoor advertisements and informal infrastructure [16]. A survey conducted in Addis Ababa, Ethiopia, revealed that 78% of residents had observed a sharp increase in outdoor advertising, while 82% believed it significantly diminished the city's visual appeal. Similarly, a study by Borowiak et al. (2024) in Gniezno, Poland, found that buildings covered in outdoor advertisements were consistently rated as less attractive by the public, even when the underlying architectural quality was high [11]. This suggests that visual clutter significantly distorts public perception of the urban environment, eroding cultural identity and aesthetic value. These findings reflect a growing recognition of visual pollution as a policy-relevant issue, yet most urban areas still lack the monitoring frameworks and data-driven tools needed to address it systematically.

Governments around the world are beginning to recognize visual pollution as a legitimate threat to urban quality of life and public well-being. In Saudi Arabia, the Ministry of Municipal and Rural Affairs and Housing (MOMRAH) has taken significant steps by formally identifying 30 types of visual pollutants, such as signage, graffiti, and damaged infrastructure, and integrating visual pollution mitigation into the country's Vision 2030 urban improvement strategy [31]. This institutional acknowledgement highlights a growing awareness that visual degradation is not merely an aesthetic issue but a barrier to sustainable and livable urban development. Such initiatives reflect the broader need for coordinated policies, monitoring mechanisms, and public engagement

to address the cumulative and often overlooked effects of visual clutter.

Historically, visual pollution assessment relied heavily on subjective and manual methods, such as opinion surveys, photographic comparisons, and field-based scoring. While these approaches provided valuable qualitative insights, they fell short in terms of consistency, scalability, and measurable benchmarks. The first structured framework to address this gap was introduced in 2019, using a standardized scorecard and calculator system to assess visual pollution in urban environments [41]. This framework was later expanded to include 39 types of visual pollution objects (VPOs), organized using a tiered weighting system to evaluate their relative impact [32]. Despite these advancements, the method continued to rely on field assessments, limiting its practical scalability. In recent years, deep learning breakthroughs have opened new possibilities, particularly through the use of object detection models that can automatically identify visual pollutants, such as billboards, garbage, and electric poles, from street imagery. However, object detection produces only bounding boxes, which may be inadequate for pollutants with diffuse or irregular shapes, such as litter, graffiti, or tangled overhead wires. In such cases, semantic segmentation offers pixel-level delineation, enabling area-based scoring and a more nuanced understanding of spatial spread. As such, exploring segmentation models represents a natural evolution in visual pollution assessment, combining automation, granularity, and scalability.

1.2 Research Questions

This study seeks to explore semantic segmentation as more fine-grained and interpretable method for detecting and quantifying visual pollution, particularly in urban environments with ambiguous or diffuse pollutant boundaries. It is guided by the following research questions:

- RQ1: How effectively can semantic segmentation models detect and delineate common forms of visual pollution at pixel-level precision?
- RQ2: What are the strengths and limitations of open-vocabulary segmentation models in detecting visual pollutants?
- RQ3: How can segmentation masks be used to develop area-based scoring methods for visual pollution?

1.3 Contributions

This study makes several contributions to the field of automated visual pollution assessment, particularly through the application of semantic segmentation techniques. The key contributions are as follows:

- **Dataset Creation:** Existing research on visual pollution has primarily focused on object detection models, with no publicly available datasets annotated for pixel-level segmentation. To enable this study, we constructed a dataset comprising both urban street-view scenes and close-up images of visual pollutants, annotated with ground-truth segmentation masks for six categories: *billboards*, *garbage*, *graffiti*, *potholes*, *construction materials*, and *tangled wires*.
- **Comparative Evaluation of Segmentation Models:** We benchmarked four widely used semantic segmentation architectures: U-Net, DeepLabV3+, Feature Pyramid Network (FPN), and SegFormer, to evaluate their effectiveness in detecting visual pollutants across complex urban scenes. In addition, we explored the potential of CLIPSeg and OVSeg, two open-vocabulary segmentation models, by analyzing their out-of-the-box generalization to visual pollution categories. In addition, we also assess the impact of fine-tuning on CLIPSeg. This comparison provides insights into both conventional and prompt-based segmentation approaches.
- **Adaptation of a Visual Pollution Quantification Framework:** Building on the Visual Pollution Index (VPI) scoring method proposed by Gupta et al. [17], we extended the framework to incorporate pixel-based outputs from segmentation models. The revised approach enables more granular, area-based scoring of visual pollutants, allowing for a more interpretable and spatially aware quantification of pollution levels.

1.4 Dissertation Structure

This dissertation is organized into seven chapters, each designed to progressively develop and support the central objectives of the study:

- **Chapter 1: Introduction:** Introduces the motivation behind the study, outlines the research problem and objectives, and summarizes the main contributions of the work.

- **Chapter 2: Related Works:** Reviews prior research on visual pollution, covering manual assessment frameworks, automated visual pollution detection, and emerging quantification techniques. It also identifies key research gaps that this dissertation aims to address.
- **Chapter 3: Data:** Describes the processes involved in constructing and annotating the dataset, including image sourcing, pseudo-label generation, and manual refinement. It also presents exploratory data analysis to highlight key characteristics and distributions.
- **Chapter 4: Methodology:** Details the semantic segmentation models used in the study, including conventional supervised models (U-Net, DeepLabV3+, FPN, and SegFormer) and an open-vocabulary model (CLIPSeg). The chapter also outlines the preprocessing pipelines, training procedures, and evaluation methods.
- **Chapter 5: Results and Analysis:** Presents the experimental setup, compares the performance of both conventional and open-vocabulary segmentation models, and includes robustness testing.
- **Chapter 6: Visual Pollution Scoring:** Introduces a pixel-based adaptation of the Visual Pollution Index (VPI) framework and demonstrates how segmentation outputs can be used to compute interpretable pollution scores.
- **Chapter 7: Conclusion:** Summarizes the key findings and contributions, discusses the limitations of the current work, and outlines directions for future research in automated visual pollution assessment.

Chapter 2

Related Works

2.1 Traditional Visual Pollution Assessment

Traditional visual pollution assessment involves a variety of methods that were developed prior to the recent advances in computer vision and deep learning. Earlier approaches to assessment relied on both quantitative and qualitative tools, including field surveys, expert judgment, and spatial analysis.

A common initial step was to measure public perceptions through field surveys, where observers rated locations according to their annoyance or acceptance of visible pollutants. For example, Chmielewski et al. combined intervisibility analysis, which measures the number and type of advertisements visible from an observer's viewpoint, with structured public surveys to produce spatial maps of perceived pollution levels [14]. These methods often used Likert scales and open-ended feedback, providing valuable but somewhat subjective insights. To address this subjectivity, researchers created audit frameworks based on expert assessment and multi-criteria decision-making techniques, such as the Analytical Hierarchy Process (AHP). For instance, Wakil et al. developed an assessment tool that assigned weights to different categories of visual pollutants based on their impact, using field checklists together with expert scoring [41]. This enabled the quantification and comparison of pollution across different urban sites.

Strategic analysis methods, such as SWOT (Strengths, Weaknesses, Opportunities, Threats) and QSPM (Quantitative Strategic Planning Matrix), have also been utilized for diagnosing sources of visual pollution and prioritizing mitigation strategies. Allahyari et al. applied these frameworks in Tehran's urban squares to systemically identify, rank, and plan intervention priorities, taking into account the perspective of planners, residents, and stakeholders [9]. With the increasing use of geospatial technologies,

quantitative GIS analyses have become more common. For instance, Karimipour et al. simulated the impact of high-rise towers on the cityscape by modeling blocked sightlines and reductions in visible landscape area, leading to more objective spatial metrics [22].

Although these traditional methods provide important foundational insights regarding visual pollution, they are typically labor-intensive, require significant manual effort, and can be influenced by the subjective bias of respondents or experts. They are difficult to scale to large or rapidly changing environments, and often focus only on single categories of pollutants, which does not account for the full complexity of modern urban scenes. Therefore, researchers have been trying to pursue more scalable, data-driven approaches. The shift towards automated and image-based techniques, particularly those that use deep learning, is addressed in the next sections.

2.2 Object Detection Models for Visual Pollution

With the increasing availability of public image datasets and the rapid development of deep learning, object detection models have become central to automated visual pollution assessment. These models, typically based on architectures such as YOLO, Faster R-CNN, Detectron2, and EfficientDet, predict the presence and bounding boxes of visual pollution objects in street-level imagery.

Recent studies have significantly contributed to the diversity of datasets and pollutant categories. For example, AlElaiwi et al. developed the large-scale MOMRAH dataset in Saudi Arabia, containing over 34,000 images classified into excavation barriers, potholes, and dilapidated sidewalks [7]. Using YOLOv5x, they achieved precision and recall scores above 0.88 and mean Average Precision (mAP) scores 0.93, demonstrating that object detection can be both practical and accurate for road-based visual pollution monitoring [8]. In Bangladesh, Hossain et al. created a dataset of 1,400 urban images with six pollution classes, achieving an mAP of approximately 0.85 using YOLOv5 [20]. Similarly, Titu et al. introduced a dataset of 5,653 images from South Asian cities, covering nine classes such as bricks, wires, towers, and street litter, and obtained 0.712 mAP using YOLOv5, also demonstrating successful deployment on edge device like Raspberry Pi for real-time detection [39]. Mazroa et al. combined YOLOv5 with explainable AI techniques in detecting pollutants across 4,400 images from Saudi Arabia, reporting mAP scores above 0.86 and classification accuracies close to 0.98 [30].

A significant milestone was achieved by Gupta et al., who linked object detection outputs with a Visual Pollution Index (VPI) calculation, quantifying urban aesthetic degradation through a combination of count-based scores and severity-based weights [17]. This linkage between detection results and composite indices has since influenced later research. Notably, Aryan Sharma's undergraduate dissertation at The University of Edinburgh expanded this framework by assembling a visual pollution dataset and employing object detection models, including YOLO and Faster R-CNN, to detect visual pollutants such as billboards, garbage, potholes, graffiti, construction materials, and wires, before computing VPI scores from these detections [38].

Despite their success in identifying the presence and approximate locations of visual pollutants, object detection models are inherently limited to producing bounding boxes. This restricts pollution assessment to coarse estimations of coverage and presence, without providing pixel-level outlines of pollutant regions. As a result, such approaches may overestimate the affected area and cannot fully capture complex, irregular shapes or overlapping objects. These limitations motivate the use of semantic segmentation methods, which can offer pixel-level precision for more accurate and interpretable pollution scoring, an approach explored in the following sections.

2.3 Semantic Segmentation Models

While object detection models are widely employed for visual pollution assessment, they are limited to predicting bounding boxes, making it difficult for more complex pollutants such as graffiti, garbage, and cluttered wires. Semantic segmentation models offer an opportunity for more interpretable visual pollution assessment, as they perform pixel-wise classification, allowing for precise identification of the spatial extent of visual pollution. This pixel-level granularity unlocks new possibilities for quantitative scoring, such as measuring the proportion of an image affected by pollutants.

Although semantic segmentation models are highly successful in biomedical imaging, urban scene analysis, and environmental monitoring, their systematic application to visual pollution detection remains limited. Prominent architectures include U-Net, DeepLabV3+, Feature Pyramid Network (FPN), and SegFormer, each delivering fine-grained, interpretable segmentations through different approaches. U-Net's encoder-decoder structure excels with limited data [37]. DeepLabV3+ uses spatial pyramid pooling to capture multi-scale context and sharp boundaries [12]. FPN exploits hierarchical features for improved segmentation across scales [27]. SegFormer employs

Transformers for efficient and robust semantic segmentation [42].

None of these models, however, have been evaluated in the context of visual pollution. To the best of our knowledge, existing literature is limited to object detection approaches, leaving the application and comparative benchmarking of semantic segmentation methods an open and important research gap. Addressing this gap, we investigate the potential of pixel-level segmentation for more accurate and interpretable assessment of visual pollution.

2.4 Open-Vocabulary Segmentation Models

Traditional and supervised semantic segmentation approaches are limited to fixed sets of categories defined in their training data. In contrast, recent advances in open-vocabulary segmentation enable models to segment objects and regions specified by arbitrary text prompts, even if those categories were unseen during training. This method presents substantial opportunities for visual pollution assessment, allowing scalable detection of novel or region-specific pollutant types without the need of exhaustive retraining. A recent comprehensive survey by Zhu and Chen provides an overview of the evolution, methodologies, and application opportunities for open-vocabulary detection and segmentation, highlighting its flexibility and adaptability for emerging visual concepts [45].

Open-vocabulary segmentation models use vision-language architectures pretrained on large image-text datasets. CLIPSeg integrates CLIP’s representations with a segmentation decoder that leverages transformer architecture, enabling segmentation from text prompts and effective zero-shot, one-shot, and referring expression tasks [28]. This approach achieves strong benchmark results, proving valuable for diverse or evolving categories. OVSeg advances this further by adapting pre-trained CLIP for mask-based classification and fine-tuning on diverse mask-category pairs via weak supervision [26]. With techniques like mask prompt tuning, OVSeg enhances masked region classification and addresses domain shifts in segmented crops.

Despite these advances, applying open-vocabulary segmentation to visual pollution remains unexplored in prior literature. Leveraging such models, as investigated in this dissertation, enables robust segmentation and quantification of visual pollution driven by flexible, interpretable text prompts. This allows practitioners to move beyond rigid pollutant definitions and rapidly adapt assessment pipelines to changing environmental context.

Chapter 3

Data

The effectiveness of visual pollution segmentation model depends on both the diversity and the representativeness of the underlying dataset. In this chapter, we describe the procedures and specifications for assembling and organizing the custom segmentation dataset used in this study. Building on earlier object detection-oriented works, this research emphasizes pixel-level annotation, which enables a more granular analysis of pollutant extent than bounding box methods [7, 17, 20, 39]. The dataset has been curated from multiple public sources, pre-processed for quality assurance, and split into training, validation, and test subsets [38].

3.1 Dataset Collection and Sources

The dataset for this research was assembled by aggregating samples from multiple publicly available sources, each targeting different types of urban visual pollutants. Broadly sampling from diverse datasets enhances both the variability and generalizability of the segmentation models, while also reducing geographic or contextual bias. This dataset builds directly on prior work which compiled and curated visual pollution imagery for object detection tasks [38]. However, this study extends this foundation by re-annotating the images for semantic segmentation, generating pixel-level masks for each pollutant class. This shift from bounding boxes to pixel-wise annotations supports more precise delineation of pollutant boundaries, enabling fine-grained analysis of their spatial extent.

The pollutant categories chosen for the study reflect common urban visual pollution types discussed in existing literature. For each class, source details are as follows:

- **Billboard:** Includes outdoor advertisements such as large billboards, signboards,

and banners. Images were collected from a publicly available Billboard Object Detection Dataset, which contains varied signage styles and covers multiple regions [33].

- **Pothole:** Urban road-surface defects, sourced from the Pothole Object Detection Dataset [13], and supplemented by road damage samples from AlElaiwi et al. [7], covering diverse surface materials, conditions, and lighting.
- **Garbage:** Litter and waste materials in streetscapes, taken from the Garbage Detection Dataset publicly available in Roboflow, containing images of trash in varied urban backgrounds [1].
- **Construction Material:** Building supplies such as bricks and sand present in the urban environment, unified from multiple classes in the Dhaka Visual Pollution Dataset publicly available on Kaggle [5].
- **Wires:** Overhead electrical and telecommunication cables sourced mainly from the Dhaka Visual Pollution Dataset [5] and additional samples from a publicly available Roboflow dataset [4], chosen for their dense urban wiring infrastructure.
- **Graffiti:** Samples of graffiti, including street art and tags, collected from publicly available Roboflow dataset [2], the Madrid Graffiti Signatures dataset [3], and a curated set described in Lavi et al. [24].
- **Negative Samples:** To reduce false positives and improve generalization, the dataset includes images with no target pollutants, representing clean urban scenes. These were collected from sources such as Google Street View and open photo datasets using keywords for pollutant-free settings (e.g., parks, tourist areas, deserts). Including negative samples helps the model learn to correctly assign background labels and distinguish pollutants from true background in diverse real-world conditions.

After consolidation and quality review, the final dataset comprises **5,981 images**, split into **4,075** for training, **603** for validation, and **1,303** for testing. By combining data sources with varied geographic origins and environmental contexts, the dataset provides a heterogeneous foundation for segmentation, ensuring that models learn to delineate pollutant boundaries and shapes across diverse settings. Representative image examples are presented in Figure 3.1.

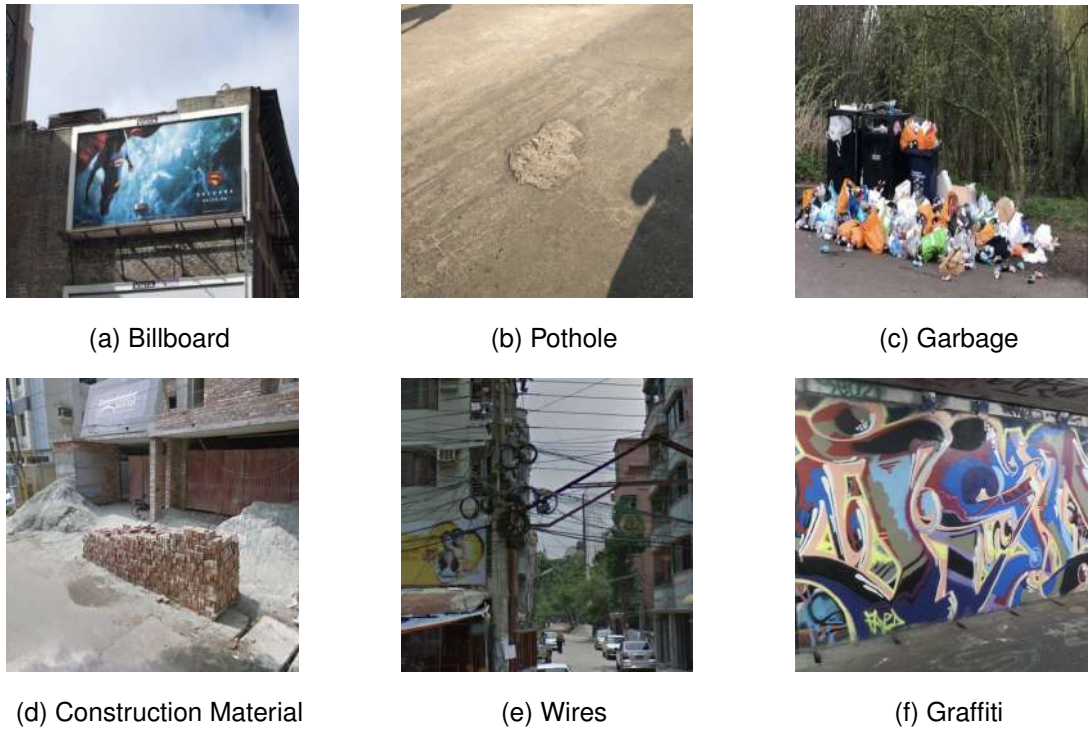


Figure 3.1: Representative examples of six pollutant classes in the segmentation dataset.

3.2 Class Distribution

In semantic segmentation, analysis of class distribution is performed at the pixel level rather than by counting object instances. Table 3.1 presents the breakdown of pixel counts and their corresponding percentages for each class, including background pixels, across the training, validation, and test splits. The distribution reveals a pronounced class imbalance: the majority of pixels are labeled as background, while classes such as construction material, wires, and pothole comprise only a small fraction of the total annotated pixels.

This imbalance is typical in urban datasets, where background naturally prevails and certain pollutants are rare or cover small regions. Left unaddressed, it biases models toward majority classes and reduces accuracy on minority ones. To mitigate this, per-class weights inversely proportional to pixel frequency were applied in the loss function, ensuring greater emphasis on under-represented classes [10].

To further improve generalization, models were initialized with pretrained weights from large-scale datasets (e.g., ImageNet, COCO). Transfer learning accelerates convergence and strengthens feature extraction, particularly for minority classes, though gains vary by architecture and domain similarity [21].

Table 3.1: Pixel distribution per class by split (percentages are within each split).

Class	Train		Validation		Test	
	Pixels	%	Pixels	%	Pixels	%
Background	1,006,680,392	72.29	150,160,791	73.44	313,880,211	72.52
Billboard	36,951,189	2.65	5,006,467	2.45	11,177,614	2.58
Construction Material	13,219,217	0.95	1,768,642	0.86	4,061,553	0.94
Garbage	84,542,092	6.07	12,304,098	6.02	29,964,930	6.92
Graffiti	210,071,680	15.09	29,740,533	14.54	63,033,028	14.56
Pothole	14,885,896	1.07	2,100,636	1.03	4,109,992	0.95
Wires	26,234,117	1.88	3,394,800	1.66	6,620,848	1.53

3.3 Pollutant Types per Image

We also examine the number of pollutant types per image, which reflects scene complexity and task difficulty. Images with multiple pollutants require the model to distinguish and localize diverse regions within a single frame. As shown in Figure 3.2, most images (over 70% per split) contain only a single pollutant class, while about 15% are clean with no pollutants. Multi-class scenes are less common, and images with three or more pollutants are rare (under 2.5%).

These results demonstrate that, for the vast majority of cases, the segmentation task involves isolating a single foreground pollutant from the background. Only a small subset of images require multi-class segmentation, and scenes with three or four pollutant types are rare (under 2.5% across all splits). The dominance of single-class images may make segmentation simpler for most samples, but it also means model exposure to genuinely multi-pollutant scenes is limited, potentially affecting generalization when faced with more complex scenarios.

3.4 Class Co-occurrence

To examine statistical relationships between pollutants, we computed class co-occurrence matrices for each data split (Figure 3.3). These heatmaps show how often pollutant pairs appear together in the same image, highlighting contextual dependencies and challenges for model learning. Most images contain only one dominant foreground class, consistent with the dataset’s real-world bias.

Certain pairs, such as *billboard–wires* and *billboard–construction material*, co-occur

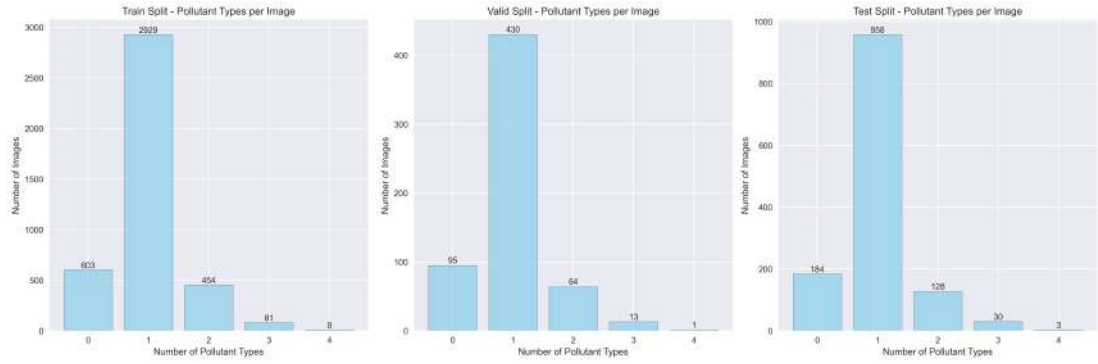


Figure 3.2: Distribution of pollutant types per image across train, validation, and test splits.

relatively often, reflecting common urban settings. In contrast, pollutants like *garbage*, *graffiti*, and *pothole* usually appear alone, suggesting that they are captured in closer views rather than broad urban scenes. Importantly, these co-occurrence patterns are consistent across the train, validation, and test sets, indicating balanced sampling.

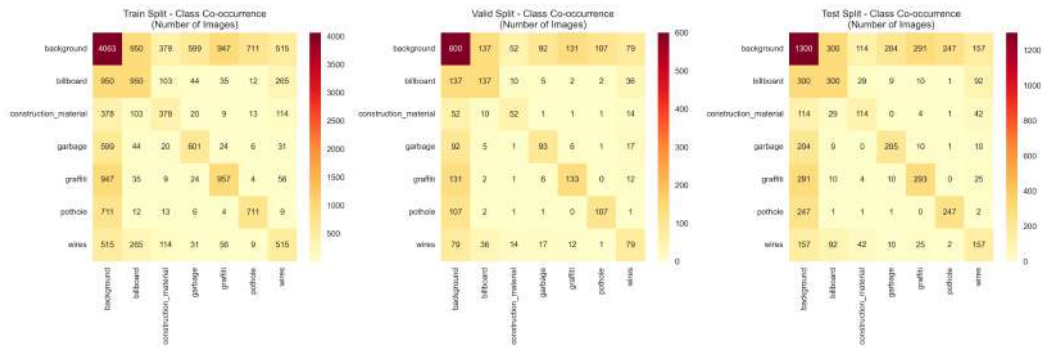


Figure 3.3: Class co-occurrence matrix.

3.5 Pollutant Coverage Analysis

We measure pollutant coverage, defined as the percentage of each image annotated as one or more pollutants (excluding background). This quantifies the extent of visual pollution per image, ranging from minor to severe cases. On average, pollutants cover about 25% of the image area across all splits, though variation is wide: some images are nearly clean, while others reach full coverage. This indicates that while most scenes include only partial pollution, there also exist close-up captures where pollutants dominate the entire frame.

To analyze pollutant types individually, Table 3.2 reports average, median, and maximum coverage for images where each pollutant appears. Graffiti and garbage typically occupy larger areas, often due to close-up captures, while potholes, wires, and construction material cover smaller regions, suggesting different scene contexts and detection challenges.

Table 3.2: Coverage by pollutant type (computed only for images where the pollutant appears).

Type	Avg (%)	Med (%)	Max (%)	Imgs
Billboard	19.99	14.35	91.33	1,391
Construction Material	13.88	10.94	65.79	545
Garbage	34.47	32.18	100.00	899
Graffiti	48.78	47.36	100.00	1,383
Pothole	8.18	4.86	56.09	1,068
Wires	15.29	9.53	86.93	753

3.6 Image Brightness Analysis

Lighting variation is an important factor in outdoor semantic segmentation, since very bright or very dark conditions can obscure features and reduce model accuracy. To measure brightness in our dataset, we examined the Value (V) channel from the HSV color space, which represents perceived brightness independently of hue and saturation.

The analysis (Figure 3.4) shows that brightness distributions across the splits are centered around a mean of 118–120, suggesting that most images are moderately lit. Only a small fraction are captured under very dark ($V < 50$) or very bright ($V > 200$) conditions. This balance supports stable training but provides limited exposure to extreme scenarios such as night scenes or harsh sunlight. To address this limitation, we applied photometric augmentations including random brightness and contrast changes, color jittering, saturation adjustments, and adaptive histogram equalization. These help the models generalize better to diverse lighting conditions.

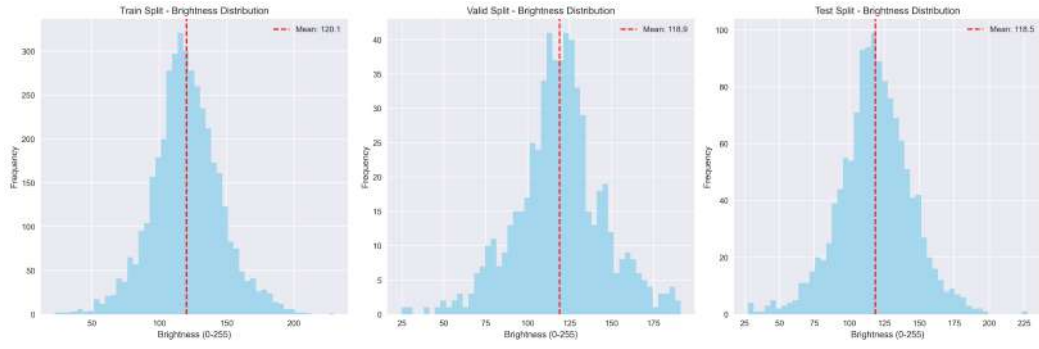


Figure 3.4: Brightness distribution histograms for training, validation, and test splits. The red dashed line indicates the mean brightness for each split.

3.7 Model-assisted Annotation and Mask Refinement

To build high-quality segmentation masks efficiently, we adopted a hybrid strategy combining model-assisted labeling with manual refinement. A subset of images was first annotated pixel-wise in CVAT to establish the training set. Several architectures (U-Net, DeepLabV3+, FPN, SegFormer) were trained and benchmarked, with SegFormer achieving the best validation results. This model was then used to generate preliminary masks (pseudo-labels) for the remaining unlabeled images, accelerating annotation while ensuring baseline accuracy.

All model-generated masks were subsequently refined in CVAT to correct missed regions, sharpen boundaries, and resolve ambiguities. This human-in-the-loop process balanced scalability with quality, following best practices in dataset curation where predictive annotation expedites large-scale labeling [36]. Examples of mask refinement are shown in Figure 3.5.



Figure 3.5: Examples of model-assisted segmentation masks before and after manual refinement. Each pair shows the original prediction (*Before*) and the refined mask (*After*).

Chapter 4

Methodology

This chapter outlines the methodological framework for urban visual pollution segmentation. We evaluate both conventional supervised models and emerging open-vocabulary approaches to delineate pollutant classes and ensure generalization across diverse urban scenes. The chapter covers model selection, training, evaluation metrics, augmentation strategies, hyperparameter tuning, and robustness testing.

4.1 Conventional Supervised Segmentation Models

To establish strong baselines for visual pollution segmentation, we evaluate four well-established supervised deep learning models: U-Net, DeepLabV3+, Feature Pyramid Network (FPN), and SegFormer. Each architecture embodies distinct design principles and innovations in semantic segmentation.

4.1.1 U-Net

U-Net is a convolutional encoder–decoder network designed for pixel-level segmentation, originally introduced for biomedical imaging tasks with limited annotated data [37]. Its “U-shaped” structure combines downsampling and upsampling paths, with skip connections preserving spatial detail and improving segmentation of small or thin objects.

We implement U-Net using the `segmentation_models_pytorch` (SMP) library with a ResNet-34 encoder pretrained on ImageNet [43]. The encoder extracts multi-scale features, which are fused with decoder layers via skip connections. The decoder progressively upsamples these features using convolutional blocks, with output mapped

to class logits by a final 1×1 convolution. Unlike the symmetric design in the original paper, our implementation uses a deeper ResNet encoder paired with a simpler convolutional decoder. This asymmetry, common in modern U-Net variants, improves feature learning while retaining the overall encoder–decoder structure. A stage-wise summary of the architecture is provided in Table A.1 in the Appendix.

4.1.2 DeepLabV3+

DeepLabV3+ extends DeepLabV3 with a decoder module for improved boundary refinement [12]. Its encoder employs Atrous Spatial Pyramid Pooling (ASPP) to capture multi-scale context using dilated convolutions, while the decoder combines high-level encoder features with low-level features to recover spatial detail.

We implement DeepLabV3+ via the `segmentation_models_pytorch` (SMP) library, using a ResNet-34 backbone pretrained on ImageNet. The decoder upsamples encoder features, merges them with low-level features, and applies 3×3 convolutions before final bilinear upsampling. Outputs are raw logits mapped to the target classes, with optional dropout for regularization.

Compared with DeepLabV3, the added decoder significantly enhances boundary segmentation. While we use ResNet-34 for efficiency, the architecture remains compatible with deeper backbones such as ResNet-101 or Xception. Figure A.2 in the Appendix visualizes the high-level architecture of DeepLabV3+.

4.1.3 Feature Pyramid Network (FPN)

Feature Pyramid Network (FPN) enhances multi-scale feature extraction by combining semantic-rich, low-resolution features with fine, high-resolution ones [27]. This design improves segmentation across object sizes, from small wires to large billboards.

FPN builds a feature hierarchy through three components: (i) a bottom-up pathway from the encoder, (ii) a top-down pathway that upsamples semantic features, and (iii) lateral connections that fuse features at each scale. The result is multi-resolution feature maps with both semantic strength and spatial precision. The original FPN design is illustrated in Figure A.3 in the Appendix section.

In our experiments, FPN is implemented using `segmentation_models_pytorch` with a ResNet-34 backbone pretrained on ImageNet. Predictions are generated from the fused pyramid features and output as logits for the loss function. Compared with

single-scale encoder–decoders, FPN explicitly exposes multi-scale features, improving robustness to object size variation.

4.1.4 SegFormer

SegFormer is a transformer-based architecture for semantic segmentation that combines strong performance with architectural simplicity [42]. It consists of two main components: a hierarchical Mix Transformer (MiT) encoder and a lightweight MLP decoder.

The MiT encoder produces multi-scale feature maps at progressively reduced resolutions ($\frac{1}{4}$ to $\frac{1}{32}$ of the input size) using overlapping patch embeddings and attention mechanisms. This design captures both local detail and global context without relying on positional encodings, enabling flexible input sizes and strong generalization across datasets.

The decoder unifies and upsamples these multi-scale features using simple MLP layers. By aggregating information from all encoder stages, it balances semantic richness with spatial precision. In our experiments, we use the MiT-B0 backbone pretrained on ImageNet, with the model outputting raw logits optimized using cross-entropy loss. An architectural overview is provided in Figure A.4 in the Appendix section.

4.2 Open-Vocabulary Segmentation Models

In addition to the conventional supervised models, we evaluate open-vocabulary segmentation approaches that are designed to generalize beyond fixed, predefined class lists. These models leverage language-guided learning and cross-modal representations to segment novel classes or ambiguous pollutant types in urban scenes. In this study, we benchmark two representative open-vocabulary segmentation frameworks, namely CLIPSeg and OVSeg, both capable of responding to textual prompts and visual–language cues. The models are first evaluated for their out-of-the-box performance on the test split, with CLIPSeg subsequently fine-tuned to further improve its results.

4.2.1 CLIPSeg

CLIPSeg is a vision–language segmentation model built on the contrastively pre-trained CLIP backbone with a lightweight transformer decoder [28, 35]. It performs binary

segmentation from free-form textual or visual prompts, making it flexible for zero-shot tasks but not directly suited to multi-class segmentation. An architecture overview is illustrated in Figure A.5 in the Appendix.

Multi-class segmentation. We adapt CLIPSeg for multi-class prediction using a prompt-wise strategy. For each pollutant type, a descriptive text prompt is supplied, producing a probability mask. Pixel-wise class labels are then assigned by selecting the highest probability across prompts, with a background class as fallback. This converts CLIPSeg’s binary outputs into a pseudo-multi-class segmentation.

Thresholding and implementation. Since CLIPSeg outputs confidence scores in $[0, 1]$, we evaluate thresholds of 0.1–0.9 and select the one yielding the highest mIoU. Our implementation uses the released ViT-B/16 encoder with a 64-dimensional decoder. Binary runs use a general urban pollutant prompt, while multi-class runs apply one prompt per pollutant type, with probability masks stacked and post-processed as above.

4.2.2 OVSeg

OVSeg is an open-vocabulary segmentation framework that extends CLIP to dense prediction [26]. It aligns pixel-level image features with text embeddings using a pixel–text contrastive learning objective, bridging the gap between global image–text alignment and per-pixel segmentation. An overview of the OVSeg architecture is presented in Figure A.6 in the Appendix.

A key innovation is its mask-adapted CLIP architecture, which incorporates mask prompt tuning to adapt representations for segmentation. During training, OVSeg learns to associate arbitrary text descriptions with spatial regions, leveraging both fully annotated and weakly supervised datasets.

In our experiments, OVSeg is applied in zero-shot mode without fine-tuning. With descriptive prompts (e.g., “*graffiti on a wall*”, “*garbage on the street*”, “*billboard*”), the model directly produces multi-class segmentation masks in a single inference step.

4.2.3 Prompt Engineering for Binary and Multi-Class Segmentation

Prompt design plays a critical role in vision–language segmentation models, particularly in open-vocabulary and zero-shot settings. In our experiments, we evaluate different prompt formulations for both binary and multi-class segmentation to analyze how natural-language phrasing influences model performance.

Binary Segmentation Prompts. For binary segmentation, CLIPSeg is tested with a single prompt describing the presence of urban visual pollution. Four variants are used, ranging from concise terms (e.g., “*visual pollution*”) to detailed descriptions that explicitly list pollutant types and contexts. These are summarized in Table 4.1.

Table 4.1: Binary segmentation prompts used for CLIPSeg experiments.

Prompt Label	Prompt Text
prompt_1	an image of urban visual pollution
prompt_2	an image showing urban visual pollution including billboards, graffiti, garbage, potholes, construction materials, and electrical wires
prompt_3	visual pollution
prompt_4	an image containing visual pollution elements such as billboards, graffiti, garbage, potholes, construction materials, or electrical wires in an urban environment

Multi-Class Segmentation Prompts. For multi-class segmentation, CLIPSeg is supplied with one prompt per pollutant class. We design four prompt sets to explore the effect of prompt complexity and contextual richness: *Concise descriptive*—uses phrases highlighting typical settings or appearances; *Detailed class-specific*—provides expanded explanations including material, location, and activity context; *Simple single-token*—contains succinct class names; *Contextual*—employs phrases emphasizing the urban context in which the pollutants appear. Table 4.2 lists the prompt variants evaluated.

Table 4.2: Multi-class segmentation prompt sets used for CLIPSeg experiments.

Prompt Label	Prompt Text
prompt_list_1 (concise descriptive)	1) an image of a billboard mounted on a building or pole; 2) an image of construction material on the ground; 3) an image of garbage or litter on the street or sidewalk; 4) an image of graffiti or spray paint on a wall; 5) an image of a pothole in the road; 6) an image of electrical wires or power lines overhead.
prompt_list_2 (detailed class-specific)	1) an image showing a billboard or advertising sign mounted on a building, pole, or billboard structure; 2) an image showing construction materials such as concrete, bricks, wood, or metal debris on the ground or construction site; 3) an image showing garbage, litter, trash, or waste materials scattered on the street, sidewalk, or urban area; 4) an image showing graffiti, street art, or spray paint markings on walls, buildings, or other surfaces; 5) an image showing a pothole, road damage, or hole in the pavement or asphalt surface; 6) an image showing electrical wires, power lines, cables, or utility lines hanging overhead or on poles.
prompt_list_3 (simple single-token)	billboard; construction material; garbage; graffiti; pothole; wires.
prompt_list_4 (contextual)	1) a billboard in an urban environment; 2) construction materials in a city setting; 3) garbage in urban areas; 4) graffiti on urban surfaces; 5) potholes in city roads; 6) electrical wires in urban infrastructure.

4.3 Data Preprocessing and Augmentations

All images and annotation masks are loaded in RGB format and resized to 256×256 pixels to ensure compatibility with model encoders. Images are normalized using ImageNet statistics to standardize input distributions across models. Masks are converted into per-pixel class labels using a predefined color-to-class index mapping required for supervised segmentation.

To enhance model generalization and robustness, a range of data augmentations is applied to the training images using the Albumentations library. These include horizontal and vertical flips, random 90° rotations, affine transformations, random resized cropping, elastic and grid distortions, coarse dropout, adaptive histogram equalization (CLAHE), color jittering, and Gaussian blur. Each augmentation is applied with fixed probabilities, as detailed in Table 4.3. Augmentations are applied only to the training set; validation and test sets are processed with resizing and normalization only. Data augmentation examples are visualized in Figure 4.1.

Table 4.3: Data augmentations applied during training and their rationale for improving robustness to varied urban scene conditions.

Augmentation	Rationale
Random horizontal & vertical flips ($p = 0.5$)	Improve invariance to scene orientation; urban objects are not always aligned.
Random 90° rotations ($p = 0.5$)	Expose model to varied viewpoints, reducing bias to fixed camera angles.
Random affine transformations ($p = 0.5$)	Simulate scale, position, and rotation changes of pollutants.
Random resized crop ($p = 0.5$)	Focus on object sub-regions; robust to size and location variation.
Elastic transform ($p = 0.2$)	Mimic real-world elastic distortions (e.g., camera shake).
Grid distortion ($p = 0.2$)	Simulate subtle warping and perspective changes.
Coarse Dropout ($p = 0.5$)	Mimic occlusions or missing data; train model to handle partial information.
CLAHE ($p = 0.5$)	Enhance local contrast; useful for varied lighting and shadows.
Color jitter ($p = 0.5$)	Vary brightness, contrast, saturation, and hue for lighting robustness.
Gaussian blur ($p = 0.2$)	Simulate mild blur or low-quality imagery.

Figure 4.1: Examples of original and augmented training images using various transformations applied during preprocessing.

4.4 Optuna Hyperparameter Search

To determine the optimal training configurations for each segmentation model, we employed Optuna, an open-source hyperparameter optimization framework that sup-

ports dynamic, model-specific search spaces, efficient exploration through advanced samplers, and automated tracking of performance metrics [6].

The search space included common training hyperparameters, including learning rate, batch size, weight decay, optimizer type, loss function, and, where applicable, dropout probability. Table 4.4 summarizes the parameters and corresponding ranges or options for each model type.

Due to time constraints, each model was optimized over 20 trials, with 5 training epochs per trial. The best hyperparameter set, selected based on the lowest validation loss, was then used for final training over 50 epochs to produce the models evaluated in subsequent experiments.

Table 4.4: Hyperparameter search space for Optuna optimization.

Parameter	Search Range / Options	Applicable Models
Learning rate	$1 \times 10^{-5} - 1 \times 10^{-3}$ (log scale)	All
Weight decay	$1 \times 10^{-6} - 1 \times 10^{-3}$ (log scale)	All
Optimizer	{AdamW, SGD}	All
Loss function	{Cross-entropy, Dice, Focal}	All
Encoder backbone	{ResNet-34}	U-Net, FPN, DeepLabV3+
	{MiT-B0}	SegFormer
Dropout probability	0.1 – 0.5	U-Net, FPN, DeepLabV3+
	None	SegFormer

4.5 Experimental Setup

Hardware & Software. Experiments were run on a University workstation with an NVIDIA RTX 2080 Ti GPU (11,GB VRAM). The codebase used Python 3.12, PyTorch 2.7, Albumentations 2.0, Optuna 4.4, and Segmentation Models PyTorch (SMP) 0.5 in a dedicated Conda environment for reproducibility.

Model Architectures. We evaluated U-Net (custom), U-Net/FPN/DeepLabV3+ from SMP with ResNet-34 encoders, and SegFormer with a MiT-B0 backbone pretrained on ImageNet. Dropout usage was tuned via Optuna.

Training Pipeline. Models were trained for 50 epochs (Optuna trials: 5 epochs) using cross-entropy loss with class weighting from pixel frequencies. Optimizers were AdamW or SGD (momentum 0.9) with ReduceLROnPlateau or CosineAnnealingLR

schedulers. Checkpoints were saved on validation-loss improvement. Metrics logged per epoch included loss, mean IoU, and Dice score.

Evaluation Protocol. Validation was performed each epoch on the held-out split. Reported results (Chapter 5) correspond to the checkpoint with the lowest validation loss.

4.6 Loss Function and Evaluation Metric

Training employed the **cross-entropy loss**, a standard choice for multi-class segmentation. For a pixel with C classes, the loss is

$$\mathcal{L}_{\text{CE}} = - \sum_{c=1}^C y_c \log(\hat{y}_c), \quad (4.1)$$

where $y_c \in \{0, 1\}$ is the ground truth indicator and $\hat{y}_c \in [0, 1]$ is the predicted probability (from softmax). This encourages high confidence for the correct class while penalizing incorrect predictions.

Model performance is evaluated using **mean Intersection over Union (mIoU)**. For class c , IoU is

$$\text{IoU}_c = \frac{|\text{Prediction}_c \cap \text{GroundTruth}_c|}{|\text{Prediction}_c \cup \text{GroundTruth}_c|}, \quad (4.2)$$

and averaged across classes as

$$\text{mIoU} = \frac{1}{C} \sum_{c=1}^C \text{IoU}_c. \quad (4.3)$$

mIoU is widely adopted in semantic segmentation since it evaluates both overlap quality and class imbalance.

4.7 Robustness Testing

We evaluate the resilience of our best-performing model, trained with the optimal hyperparameters found by Optuna search, to common forms of data corruption. Perturbations simulate sensor noise, defocus, lighting variation, occlusion, and transmission errors. Each perturbation type is tested at multiple severity levels to simulate progressively more challenging conditions. The full details of the perturbations and their rationale, in addition to the perturbed image examples, are presented in Table A.2 and Figure A.8 in the Appendix.

Chapter 5

Results and Analysis

This chapter presents the experimental results obtained from the methodologies and configurations described in Chapter 4. We begin by evaluating the performance of conventional supervised segmentation models, followed by an assessment of open-vocabulary approaches including CLIPSeg and OVSeg. Both quantitative and qualitative visualizations are provided to offer a comprehensive perspective on each model’s segmentation capabilities. For CLIPSeg and OVSeg, additional experiments investigate the influence of prompt formulation on segmentation accuracy. We then examine the robustness of the best performing model under a variety of image perturbations designed to simulate real-world conditions.

5.1 Conventional Supervised Models

This section evaluates the performance of the conventional supervised segmentation models introduced in Chapter 4, namely U-Net, FPN, DeepLabV3+, and SegFormer. Each model was trained using the optimal hyperparameters identified through Optuna search, and evaluated on the held-out test set.

5.1.1 Final Hyperparameters from Optuna

Before presenting the experimental results, we summarize the optimal hyperparameters identified through the Optuna search described in Chapter 4. Each configuration was selected based on the lowest validation loss after 5 training epochs during hyperparameter optimization. The selected settings were then used for full training over 50 epochs to produce the final models evaluated in this chapter. The hyperparameter details are

presented in Table 5.1.

Table 5.1: Final hyperparameters for each segmentation model, as determined by Optuna search.

Hyperparameter	U-Net	DeepLabV3+	FPN	SegFormer
Learning rate	5.30×10^{-5}	7.01×10^{-5}	9.54×10^{-4}	2.27×10^{-4}
Weight decay	1.01×10^{-6}	8.04×10^{-5}	9.72×10^{-4}	6.23×10^{-5}
Use dropout	True	True	True	False
Dropout prob.	0.105	0.240	0.100	—
Optimizer	AdamW	AdamW	SGD	AdamW

All models were trained for **50 epochs** with a **batch size of 16**, using the **ReduceLROnPlateau** scheduler and **cross-entropy loss**.

5.1.2 Model Comparison

Figure 5.1 illustrates the training and validation curves for loss and mIoU across 50 epochs for the four conventional supervised segmentation models. All models demonstrated rapid convergence within the first 10 epochs. UNet achieved the fastest convergence, reaching 90% of its best IoU in just 4 epochs, followed by DeepLabV3+ in 6 epochs, and both FPN and SegFormer in 7 epochs. SegFormer achieved the highest validation IoU (0.741), closely followed by UNet (0.738) and DeepLabV3+ (0.736), while FPN trailed slightly (0.716). The train-validation gap remained relatively small for all models, indicating effective regularization.

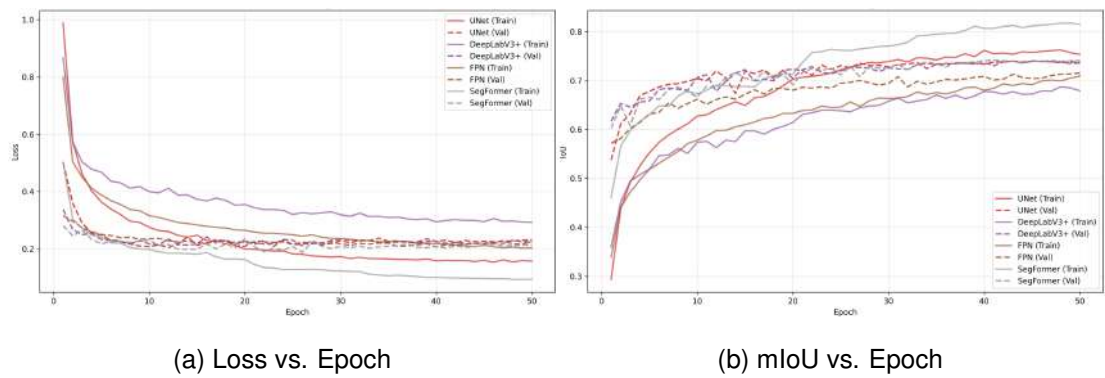


Figure 5.1: Training process analysis for conventional supervised models (UNet, FPN, DeepLabV3+, SegFormer). Solid lines indicate training; dashed lines indicate validation.

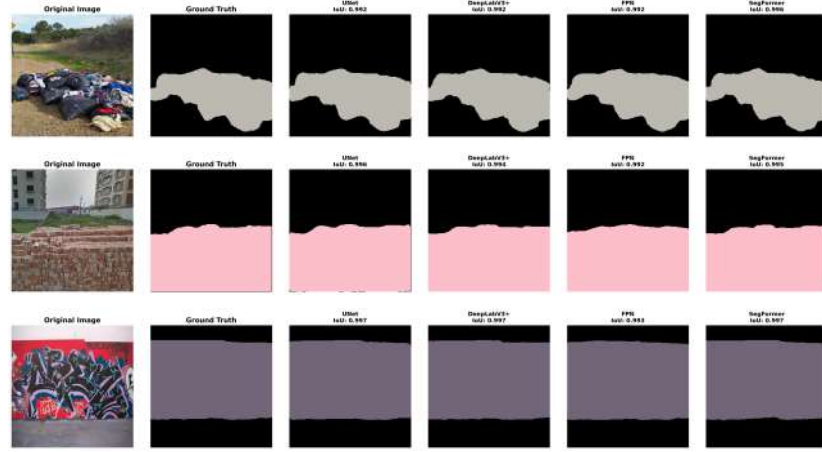
Table 5.2 presents the per-class IoU scores for each supervised model on the test set. The dominant background class achieved consistently high IoU scores across all models, reflecting its prevalence in the dataset. For urban pollutant classes, performance varied by architecture. SegFormer achieved the highest mean IoU (0.734), showing consistent improvements over other models in most classes, particularly *Garbage* (0.809), *Graffiti* (0.813), *Pothole* (0.576), and *Wires* (0.529). DeepLabV3+ followed closely with a mean IoU of 0.728. For pollutant types where images typically contain a single, clearly defined class (e.g., *Garbage*, *Graffiti*, *Construction Material*), models generally performed well with IoU values around 0.80. However, for more challenging classes where objects exhibit greater complexity (e.g., *Pothole* and *Wires*), performance did not exceed 0.60 IoU. This observation will be further supported by visual examples of best- and worst-case predictions.

Table 5.2: Per-class IoU performance on the test set for supervised models. Best result per class is highlighted in **bold**.

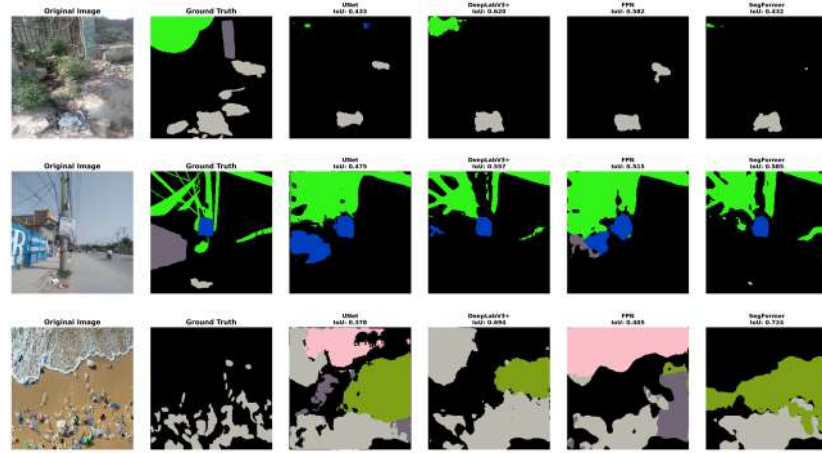
Model	mIoU	Background	Billboard	Construction	Garbage	Graffiti	Pothole	Wires
U-Net	0.7114	0.9079	0.7299	0.7449	0.7912	0.7974	0.5216	0.4870
DeepLabV3+	0.7280	0.9132	0.7242	0.7917	0.8009	0.7942	0.5602	0.5119
FPN	0.7084	0.9065	0.7153	0.7349	0.7968	0.8068	0.5154	0.4831
SegFormer	0.7340	0.9154	0.7278	0.7679	0.8088	0.8132	0.5756	0.5293

To complement the quantitative evaluation, qualitative examples are provided to illustrate the strengths and limitations of the supervised segmentation models. Two types of results are shown: best-case predictions, where all models accurately segment the target pollutants, and worst-case predictions, where performance is poor, often for challenging classes such as pothole and wires or in scenes with multiple overlapping pollutants. Figure 5.2 illustrates the best and worst-case predictions.

Across the qualitative results, several trends emerged regarding class-specific segmentation performance. Images that were close-up and contained one or two well-defined pollutants, such as billboards, garbage, or graffiti, were segmented with higher accuracy. In contrast, scenes with multiple overlapping pollutants posed greater challenges for all models. Wires were particularly problematic, as they often appeared in cluttered scenes and, due to their thin structure, tended to blend into the background, especially when positioned in front of textured surfaces like building facades. Objects with straight lines, such as scaffoldings or architectural edges, were also frequently misclassified as wires. Scattered garbage proved more difficult to segment due to its



(a) Best-case scenarios



(b) Worst-case scenarios

Figure 5.2: Qualitative examples of (a) best-case and (b) worst-case segmentation results across the four supervised models.

irregular distribution and varied appearance. Pothole segmentation was hindered by several factors in the MOMRAH dataset, from which the majority of pothole images were sourced: many were captured in low-light conditions, a significant number depicted small or visually subtle potholes, and the definition of what constituted a pothole was sometimes ambiguous, leading to inconsistent labeling in the original dataset.

5.1.3 Robustness Testing

We evaluated the robustness of the best-performing supervised model, SegFormer, under a range of perturbations applied at increasing severity levels (for details, see Table A.2 in the Appendix). Baseline performance was $mIoU = 0.734$. Table 5.3 reports the

lowest mIoU observed for each perturbation and the relative degradation compared to baseline.

Table 5.3: SegFormer robustness results based on mIoU. Percentage drop is calculated relative to the baseline (0.7340).

Perturbation	Min mIoU	% Drop
Salt & Pepper	0.3077	58.08%
Contrast Decrease	0.4456	39.29%
Gaussian Noise	0.6313	13.99%
Gaussian Blur	0.6587	10.26%
Occlusion	0.7088	3.43%
Brightness Decrease	0.7197	1.94%
Brightness Increase	0.7266	1.01%
Contrast Increase	0.7315	0.35%

The results show that Salt & Pepper noise is the most damaging perturbation, reducing mIoU by 58.08%. This reflects extreme vulnerability to high-frequency pixel corruption that disrupts boundaries. The next most severe effect comes from contrast decrease (-39.29%), simulating low-visibility conditions such as haze or fog.

Moderate performance drops occur under Gaussian noise (-13.99%) and Gaussian blur (-10.26%), indicating reliance on fine texture and edge information. In contrast, brightness shifts, occlusion, and contrast increase produce only minor changes, demonstrating resilience to illumination variations and partial object loss.

Overall, SegFormer is robust to most natural perturbations but remains highly sensitive to severe noise and reduced contrast. These findings emphasize the need for robustness-oriented training, particularly with noise- and fog-based augmentations, to prepare models for deployment in uncontrolled urban settings.

5.2 Open-Vocabulary Models

This section examines the performance of open-vocabulary segmentation models on the visual pollution segmentation task. Unlike conventional supervised models, which are restricted to the set of classes seen during training, open-vocabulary approaches leverage vision-language models to segment objects based on natural language prompts. We evaluate two open-vocabulary models: CLIPSeg and OVSeg. For CLIPSeg, we analyze

both its out-of-the-box performance and the effect of prompt engineering for binary and multi-class segmentation, followed by results from model fine-tuning. For OVSeg, we report out-of-the-box performance and prompt engineering results for multi-class segmentation.

5.2.1 CLIPSeg

As previously described in Chapter 4, CLIPSeg is a vision-language model designed for open-vocabulary segmentation, enabling pixel-level predictions guided by natural language prompts. In this study, it was evaluated in both binary and multi-class segmentation settings to assess its zero-shot capabilities for detecting urban visual pollutants.

Binary Segmentation. For the binary segmentation evaluation, CLIPSeg was tested with four prompts (Table 4.1) across nine probability thresholds ranging from 0.1 to 0.9. The IoU–threshold curves (Figure 5.3a) and IoU standard deviation–threshold curves (Figure 5.3b) illustrate the sensitivity of predictions to both prompt formulation and threshold selection.

The highest mean IoU (0.480) was achieved by Prompt 1 (*"an image of urban visual pollution"*) at a threshold of 0.50. The most consistent predictions (lowest IoU standard deviation) came from Prompt 3 (*"visual pollution"*) at a threshold of 0.20, although with a lower mean IoU of 0.337. Prompt engineering had a notable impact, with moderately descriptive prompts generally outperforming shorter ones, while overly long descriptions, such as Prompt 4, showed diminished performance. Threshold sensitivity analysis revealed substantial IoU variation, especially for longer descriptive prompts, underscoring the need for prompt-specific threshold tuning.

Multi-class Segmentation. Zero-shot multi-class segmentation was evaluated for six foreground pollutant classes using four prompt lists (Table 4.2), each tested over nine probability thresholds. As summarized in Figures 5.4a and 5.4b, prompt formulation and threshold choice again proved critical. The best overall results followed from detailed class-specific prompts (Prompt List 2) at threshold 0.50 (mean IoU 0.636), followed by concise descriptive prompts (Prompt List 1, mean IoU 0.616). Simple, single-token prompts (Prompt List 3) produced the lowest scores and least consistent predictions, while contextual prompts (Prompt List 4) offered intermediate performance. Overall, moderately descriptive, class-specific prompts outperform both minimal and overly generic phrasing, and a threshold sweet spot around 0.50 - 0.60 consistently

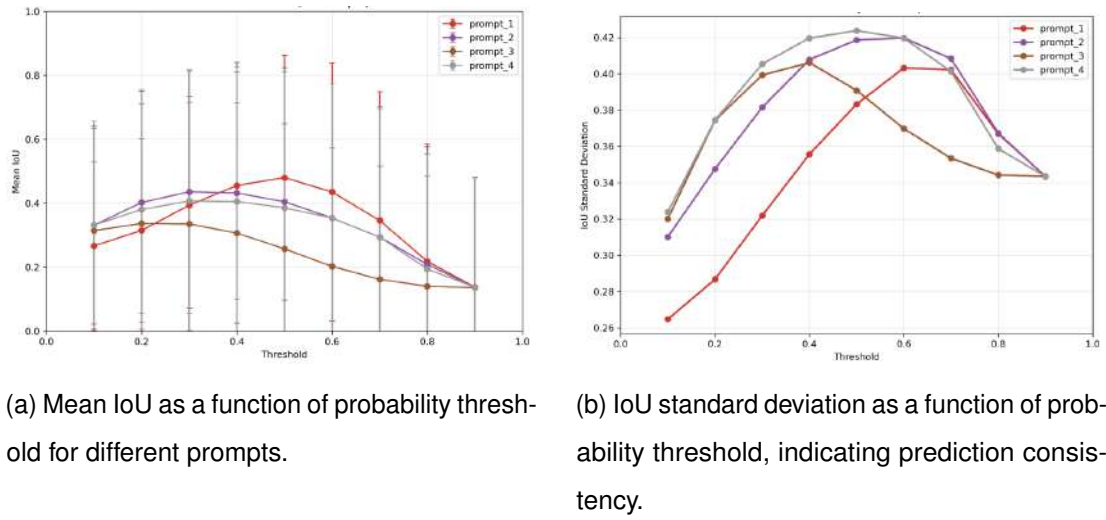


Figure 5.3: Binary segmentation performance of CLIPSeg for four different prompts across nine probability thresholds.

emerges.

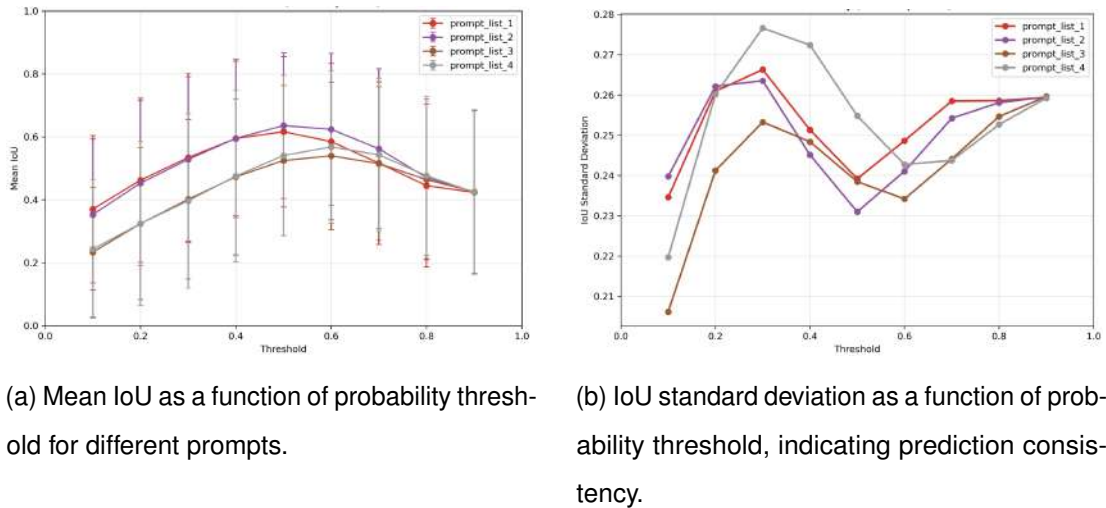


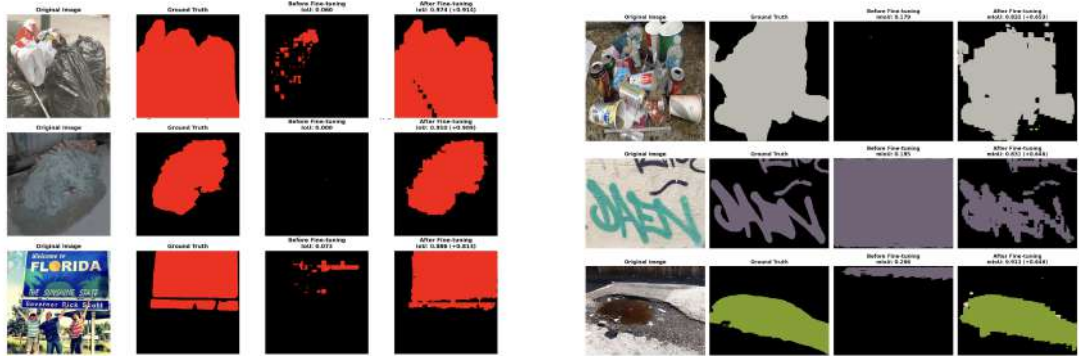
Figure 5.4: Multiclass segmentation performance of CLIPSeg for four different prompts across nine probability thresholds.

Fine-Tuning CLIPSeg. To further enhance CLIPSeg performance, we conducted fine-tuning experiments for both binary and multi-class segmentation tasks. An Optuna search was used to identify optimal hyperparameters and unfreezing strategies. Both models were trained for 50 epochs using the best parameter settings found.

For **binary segmentation**, the optimal configuration included a learning rate of 0.0006, batch size of 8, AdamW optimizer, step scheduler, and unfreezing only the decoder. This led to a substantial increase in test set mean IoU from 0.480 (zero-shot)

to 0.707 (+0.227). Meanwhile, for **multi-class segmentation**, the best results were obtained with a learning rate of 0.0005, batch size 8, AdamW optimizer, step scheduler, and unfreezing the decoder along with the last four transformer layers (totaling about 19.6% of the model’s trainable parameters). Fine-tuning increased mIoU from 0.636 to 0.781 (+0.146). The largest relative gains were observed in construction material (+0.510), pothole (+0.386), and billboard (+0.229), all of which benefited substantially from additional training. Smaller but notable improvements were achieved for wires (+0.133), garbage (+0.105), and graffiti (+0.085). While these gains indicate a clear benefit from fine-tuning, some classes, particularly pothole and wires, still exhibit relatively low absolute IoU scores (0.416 and 0.399 respectively), suggesting they remain challenging to segment. The full per-class results are provided in Appendix Table A.4.

Figure 5.5 shows before-and-after examples for both binary and multi-class segmentation, highlighting significant improvements. However, due to the thresholding step in open-vocabulary segmentation, some object boundaries remain rough, with occasional background pixels appearing inside objects when probabilities fall below the threshold.



(a) Binary segmentation examples.

(b) Multi-class segmentation examples.

Figure 5.5: Qualitative examples showing segmentation improvements from fine-tuning for CLIPSeg. For each example, columns show: *Original Image*, *Ground Truth*, *Before fine-tuning*, and *After fine-tuning*.

5.2.2 OVSeg

OVSeg was evaluated in zero-shot mode on the visual pollution dataset using the same four prompt lists as CLIPSeg, with probability thresholds ranging from 0.1 to 0.9. The best overall performance was obtained with Prompt List 3 (simple, single-token

prompts) at a confidence threshold of 0.3, yielding a macro-averaged mIoU of 0.493. Threshold analysis revealed a sharp drop in performance at very low thresholds (e.g., 0.159 at 0.1), followed by a plateau above 0.3. This behaviour is likely due to most predicted pixels being classified as background when the threshold is too high, as the model’s confidence for pollutant pixels rarely exceeded these values (see Figure A.12 in the Appendix).

Per-class results (at the best prompt–threshold setting) showed graffiti (0.542 mIoU) and billboard (0.524 mIoU) as the strongest categories, followed by garbage (0.488) and construction material (0.448). In contrast, pothole (0.107) and wires (0.191) remained the most challenging, consistent with results from supervised and CLIPSeg models. Across all prompt lists, Prompt List 3 slightly outperformed the others, though differences were marginal (≤ 0.024 mIoU). Given the complexity of fine-tuning OVSeg, no further optimization was attempted in this study, and fine-tuning experiments were focused exclusively on CLIPSeg. Examples of OVSeg predictions are shown in Figure 5.6.

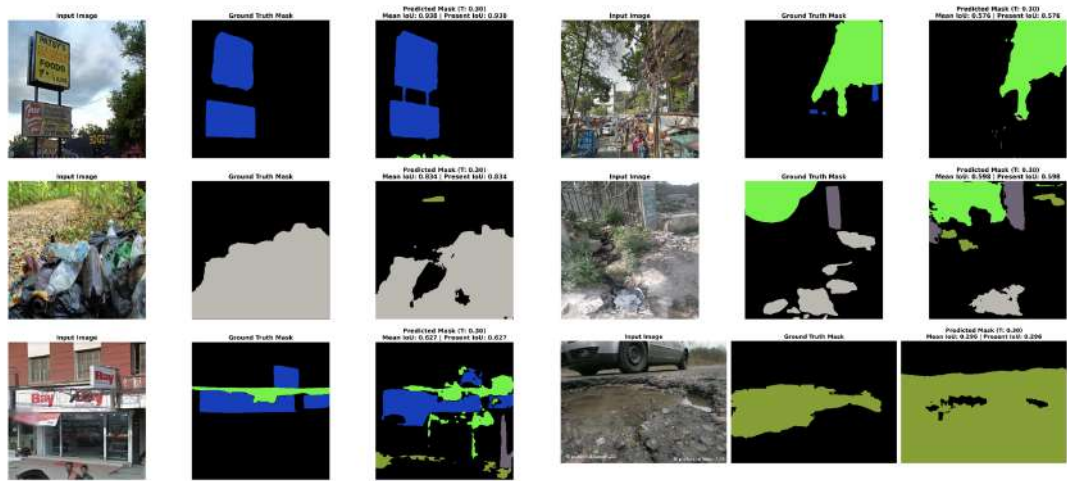


Figure 5.6: Qualitative examples of OVSeg segmentation predictions.

5.3 Discussion

5.3.1 Supervised vs. Open-Vocabulary Models

The results highlight a clear trade-off between conventional supervised models and open-vocabulary approaches. Supervised models, especially transformer-based architectures such as SegFormer, consistently achieved the highest accuracy when trained with

pollutant-specific annotations, reaching an overall mIoU of 0.734 on the test set. These models excelled on pollutant types with clear shapes and sufficient representation in the dataset, such as *garbage*, *graffiti*, and *billboards*. However, they struggled with *potholes* and *wires*, which are inherently small, irregular, and often visually ambiguous. The reliance on closed training categories further limits their generalization to unseen pollutants or domain shifts.

Open-vocabulary models, in contrast, bring flexibility by leveraging vision–language pretraining on large-scale datasets (e.g., CLIP’s 400M image–text pairs) [35]. This enables zero-shot transfer to pollutant types never explicitly annotated in training. However, their outputs often suffer from rougher boundaries and sensitivity to probability thresholds, which can fragment objects or miss fine details. OVSeg in particular showed performance plateauing above a threshold of 0.3, suggesting that pollutant regions were predicted with low confidence and easily reclassified as background. Both CLIPSeg and OVSeg also shared the same weaknesses as supervised models. Pollutants such as *potholes* and *wires* remained the hardest categories, indicating intrinsic challenges in visual distinctiveness rather than model-specific limitations.

5.3.2 The Role of Prompt Engineering

A consistent finding across experiments was the strong influence of prompt formulation. For CLIPSeg, the effectiveness of prompts depended on whether the task was binary or multi-class. In the binary case, where the goal was to separate pollutants from background, broad prompts such as “*an image of urban visual pollution*” consistently outperformed highly descriptive alternatives. These generalized prompts better captured the heterogeneous appearance of pollutants, yielding higher IoU and more stable performance. By contrast, in the multi-class setting, where the task required distinguishing between pollutant categories, class-specific prompts proved more effective. Detailed descriptions such as “*an image showing a billboard or advertising sign mounted on a building*” or “*an image showing construction materials such as concrete, bricks, wood, or metal debris on the ground or construction site*” helped the model disambiguate visually similar classes and improved category-level segmentation. This contrast highlights the need to adapt prompt design not only to the model architecture but also to the task formulation.

OVSeg, on the other hand, exhibited different behavior. Concise, single-token prompts were generally most effective, which aligns with its design: text embeddings

are averaged across short template phrases (e.g., “*a photo of a* ” [26]. Longer or more descriptive inputs did not provide additional benefit, and in some cases reduced performance due to mismatched embeddings. The divergence between CLIPSeg and OVSeg demonstrates that prompt engineering is a central component of open-vocabulary segmentation. In practice, this means that both threshold tuning and systematic prompt selection must be tailored to each model and task to achieve reliable segmentation outcomes.

5.3.3 Robustness and Deployment Considerations

Beyond accuracy on clean datasets, robustness tests emphasized the practical challenges of deploying segmentation models in uncontrolled urban environments. For SegFormer, Salt & Pepper noise caused the steepest performance drop (-58% mIoU), simulating severe sensor errors or corrupted transmissions. Reduced contrast (haze, fog, poor lighting) also had a substantial impact, while brightness shifts and mild occlusions were less damaging. These results suggest that models are more vulnerable to degradations that obscure structural boundaries than to global intensity changes.

For real-world deployment, this implies two directions. First, training pipelines should incorporate robustness-oriented augmentations, especially noise and fog simulations. Second, dataset diversity must be expanded: most current visual pollution datasets underrepresent conditions such as night scenes, strong occlusion, or region-specific urban styles. As noted by Gupta et al. [17], drone imagery can introduce new perspectives and environmental variations that complement street-level datasets. Ultimately, bridging the gap between controlled benchmarks and complex urban realities will require not only stronger model architectures but also richer datasets and adaptive training strategies such as domain adaptation or continual fine-tuning.

Chapter 6

Visual Pollution Scoring

This chapter presents the methodology used to quantify visual pollution based on semantic segmentation outputs. The work adapts and extends the Visual Pollution Index (VPI) computation framework introduced in previous studies within an object detection context [17, 38]. To make the framework compatible with segmentation outputs, instance counts are replaced with pixel-based measures. The adapted approach evaluates visual pollution in a given region or group of images by considering pollutant severity, diversity, and occurrence frequency.

6.1 Scoring Framework Adaptation

The resulting Visual Pollution Index (VPI) is defined as:

$$\text{VPI} = (W_s \cdot \text{SWS} + (1 - W_s) \cdot \text{CCR}) \cdot \text{PR} \quad (6.1)$$

where W_s is the severity weight (set to 0.7), SWS is the Severity-Weighted Score, CCR is the Counting Categories Ratio, and PR is the Pollution Ratio.

In this study, the VPI is calculated for a group of images (e.g., all Street View samples from a specific region) rather than for individual images. This design choice ensures a more reliable assessment of pollutant diversity. When applied to single images, the CCR is often low, since most images contain only one or two pollutant types, while complex scenes with multiple types are rare. Aggregating results across images captures the broader variety, severity, and frequency of pollutants in a region.

6.1.1 Severity-Weighted Score (SWS)

The SWS quantifies the relative visual impact of pollutants by combining their assigned severity values with their proportional pixel coverage across the aggregated image set. Unlike the original object detection framework, which used the number of detected instances, our segmentation-based approach measures pollutant extent by pixel area per category.

Severity values are derived from the expert-based rankings of Wakil et al. [41], obtained via the Analytical Hierarchy Process (AHP) and normalised to the range [0, 1]. In this scheme, garbage and construction material have the highest weight (1.0), followed by billboards (0.8), wires (0.6), graffiti (0.4), and potholes (0.2). The adapted formula is:

$$\text{SWS} = \frac{\sum (\text{Severity Value} \times \text{Pixel Count}_{\text{category}})}{\text{Total Pixels of All Categories}} \quad (6.2)$$

where the severity value reflects the perceived visual disturbance of each pollutant type, and the pixel count is the total number of pixels for that category across the region.

6.1.2 Counting Categories Ratio (CCR)

The CCR measures the diversity of pollutant types present in the aggregated image set. In the original object detection framework, this metric was computed by counting the number of distinct categories with at least one detected instance in a region, divided by the total number of possible categories. In our segmentation-based adaptation, detection is defined as having at least one pixel assigned to the category in any of the images within the group.

The rationale for including CCR is that a more diverse set of pollutant types generally indicates a higher visual clutter and a greater impact on the perceived quality of the environment. While a single image may only contain one or two categories, aggregating across a region allows the CCR to reflect a more representative measure of pollutant diversity. The adapted formula is:

$$\text{CCR} = \frac{\text{Number of Distinct Categories with } > 0 \text{ pixels}}{\text{Total Number of Categories}} \quad (6.3)$$

6.1.3 Pollution Ratio (PR)

The PR measures how frequently pollutants occur across the image set for a given region. A region where pollutants are consistently visible across most sampled locations

is likely to have a greater negative visual impact than a region where pollutants are rare, even if the latter has a few severe cases.

In our segmentation-based adaptation, an image is considered polluted if at least one non-background pixel is detected for any pollutant category. PR is then calculated as the fraction of polluted images over the total number of images for the region. The formula is:

$$PR = \frac{\text{Number of Images with Any Pollutant}}{\text{Total Number of Images}} \quad (6.4)$$

6.2 Example Application

To demonstrate the segmentation-based Visual Pollution Index (VPI) framework, we applied it to a real-world case study using Google Street View imagery. This choice ensures that pollutants are observed in natural urban contexts rather than in close-up or atypical viewpoints, which are common in our original dataset. The original test set was deemed unsuitable for this experiment, as many images depict pollutants in unrealistic framing, potentially leading to artificially high VPI scores due to the pollutant dominating the image frame. The methodology operates on aggregated image groups representing specific regions, enabling pollutant severity, diversity, and frequency to be assessed at a city or district level.

6.2.1 Data Collection

Dhaka, Bangladesh, was selected as a test case due to its high population density and prevalence of urban visual clutter [20]. We sampled 1,000 random coordinates from OpenStreetMap within Dhaka's administrative boundary, retrieving four Google Street View images per location (0°, 90°, 180°, 270°), for a total of 4,000 images. The best-performing segmentation model (SegFormer, see Chapter 5) was then applied to generate pixel-wise masks of six pollutant categories, which were subsequently aggregated to compute the VPI score for the region.

6.2.2 Results and Discussion

Figure 6.1 shows representative examples from Dhaka with the corresponding SWS, CCR, PR, and VPI scores. For Dhaka, the computed values were: SWS = 0.644, CCR = 1.000, PR = 0.720, resulting in an overall VPI of 0.540. In these samples, billboards and

overhead wires are the most frequently segmented categories, while garbage, graffiti, and potholes appear less often. This reflects the urban landscape of Dhaka, where signage and electrical infrastructure are visually dominant. As a comparison, Dhaka records a higher VPI (0.540) among other cities tested: Jakarta (0.467), New York (0.266), and Barcelona (0.079). These differences illustrate how the adapted scoring framework captures cross-city variations, with Dhaka and Jakarta showing greater prevalence of visual clutter compared to New York and Barcelona.

Nevertheless, these results should be regarded as proof of concept rather than definitive measurements. Because sampling relied on randomly drawn coordinates from OpenStreetMap, outcomes may vary with different sets of sampled points. Model performance also plays a critical role: low VPI values may reflect genuinely low visual pollution, but they could also result from segmentation errors or missed detections. Domain shift is another limitation, since the models were trained on mixed sources rather than exclusively on Street View, which may reduce accuracy in this context. Moreover, the absence of ground truth annotations for these test images prevents direct validation, so reported scores should be interpreted with caution.



(a) Sample 1 — SWS = 0.65, CCR = 0.50, PR = 0.90, VPI = 0.54



(b) Sample 2 — SWS = 0.71, CCR = 0.67, PR = 0.90, VPI = 0.62

Figure 6.1: Two example sets of randomly grouped images from Dhaka with corresponding segmentation predictions and VPI results.

Chapter 7

Conclusion

This dissertation explored semantic segmentation as an approach for automated detection and quantification of urban visual pollution. To address the limitations of object detection methods, we constructed a pixel-level annotated dataset covering six major pollutant categories. This dataset enabled comprehensive benchmarking of both conventional supervised models and newer vision-language frameworks.

(RQ1) Our results show that supervised segmentation models, with SegFormer as the best performing model, deliver strong results when pollutant boundaries are clear and sufficient training data is available. However, these models are restricted to fixed training categories, face challenges in generalizing to new domains, and show less than optimal performance for certain pollutant classes. **(RQ2)** Open-vocabulary models such as CLIPSeg and OVSeg provide greater flexibility, allowing prompt-based segmentation to predict new pollutant types without retraining. These models, however, are sensitive to prompt choices and probability thresholds, and their outputs sometimes lack precise object boundaries. **(RQ3)** By adapting the Visual Pollution Index (VPI), we demonstrated how segmentation-derived metrics can be aggregated into a composite score, enabling comparative assessment across cities such as Dhaka, Jakarta, New York, and Barcelona.

Despite these advances, several challenges remain. The current dataset has limited diversity, which affects robustness, especially for rare pollutant types and scenes taken under difficult conditions such as low light or in cities with distinct visual characteristics. Expanding the annotated dataset, improving training strategies, and employing domain adaptation will be important for practical deployment. Future work should also address the lack of ground truth labels for large urban datasets and ensure images are collected in a standardized manner.

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Appendix A

Supplementary Information

A.1 Model Architectures

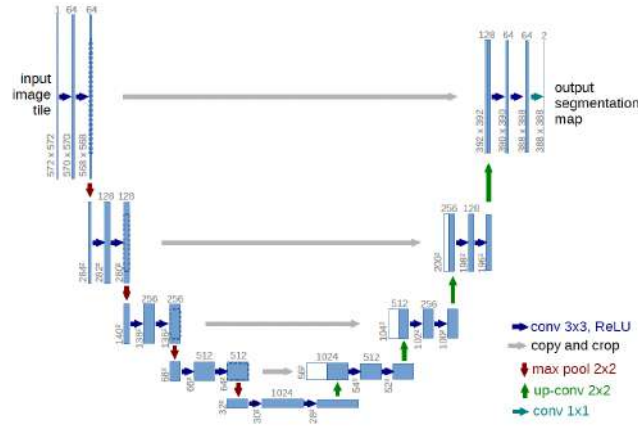


Figure A.1: The original U-Net architecture, reproduced from [37].

Table A.1: High-level schematic of the U-Net (SMP) with ResNet-34 encoder and decoder channels (256, 128, 64, 32, 16). C is the number of output classes.

Stage	Res.	Encoder Ch.	Decoder Ch.	Operation
E0	$1\times$	64	–	Stem conv + BN + ReLU
E1	$1/2\times$	64	–	ResNet block
E2	$1/4\times$	128	–	ResNet block
E3	$1/8\times$	256	–	ResNet block
E4	$1/16\times$	512	–	ResNet block
D4	$1/8\times$	–	256	Upsample + concat E3 + conv
D3	$1/4\times$	–	128	Upsample + concat E2 + conv
D2	$1/2\times$	–	64	Upsample + concat E1 + conv
D1	$1\times$	–	32	Upsample + concat E0 + conv
Head	$1\times$	–	C	1×1 conv (logits)

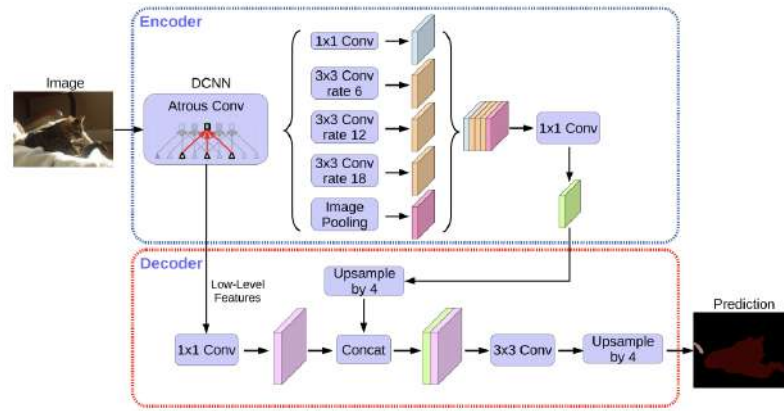


Figure A.2: High-level architecture of DeepLabV3+, reproduced from [12].

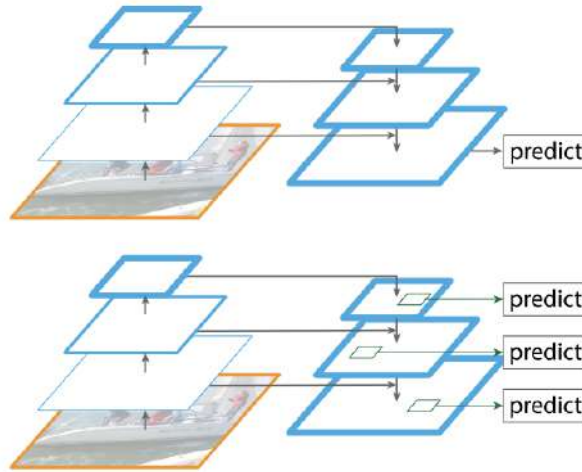


Figure A.3: Comparison between a standard top-down architecture with skip connections (top) and the Feature Pyramid Network (FPN) approach (bottom), which makes independent predictions at each pyramid level. Reproduced from [27].

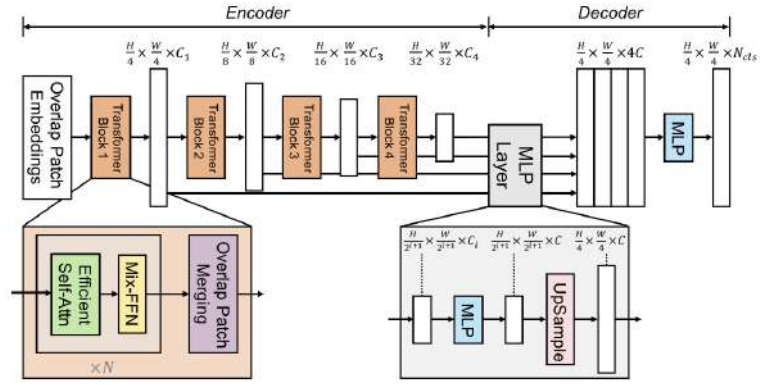


Figure A.4: The SegFormer framework consists of two main modules: a hierarchical Transformer encoder to extract coarse and fine features, and a lightweight All-MLP decoder to fuse multi-level features and predict the semantic segmentation mask. Reproduced from [42].

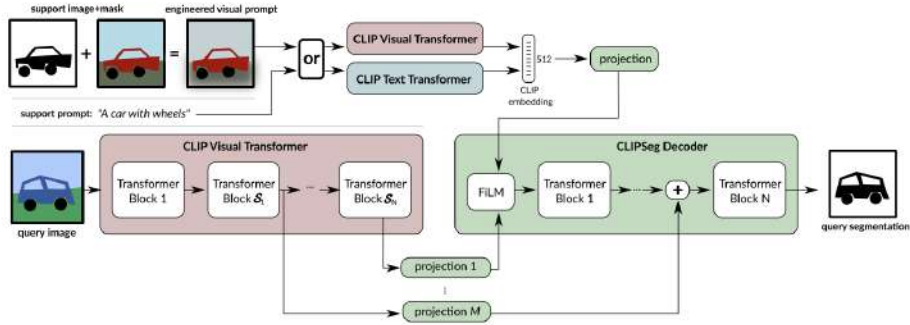


Figure A.5: CLIPSeg architecture. The model extracts CLIP embeddings from a visual prompt or text prompt, projects these representations, and uses a transformer-based decoder to produce per-pixel segmentations for a query image. Reproduced from [28].

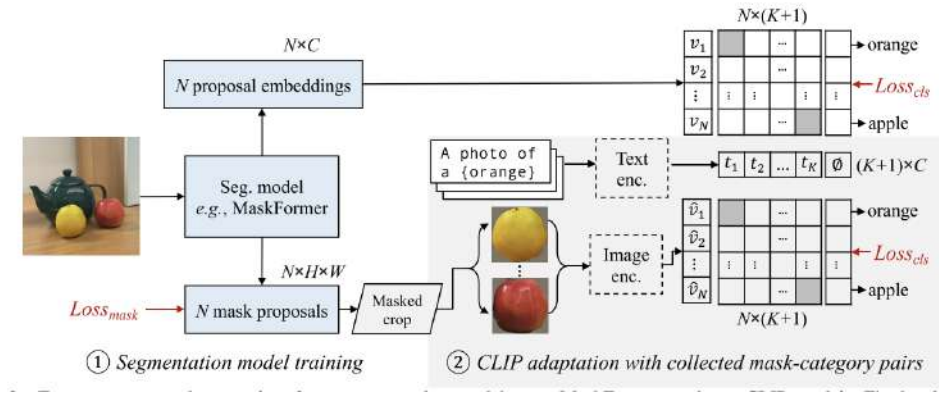
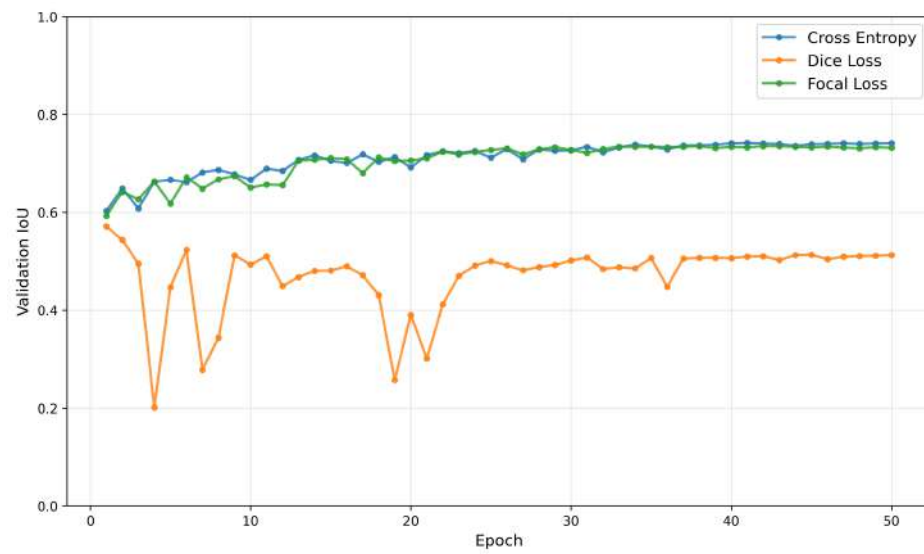
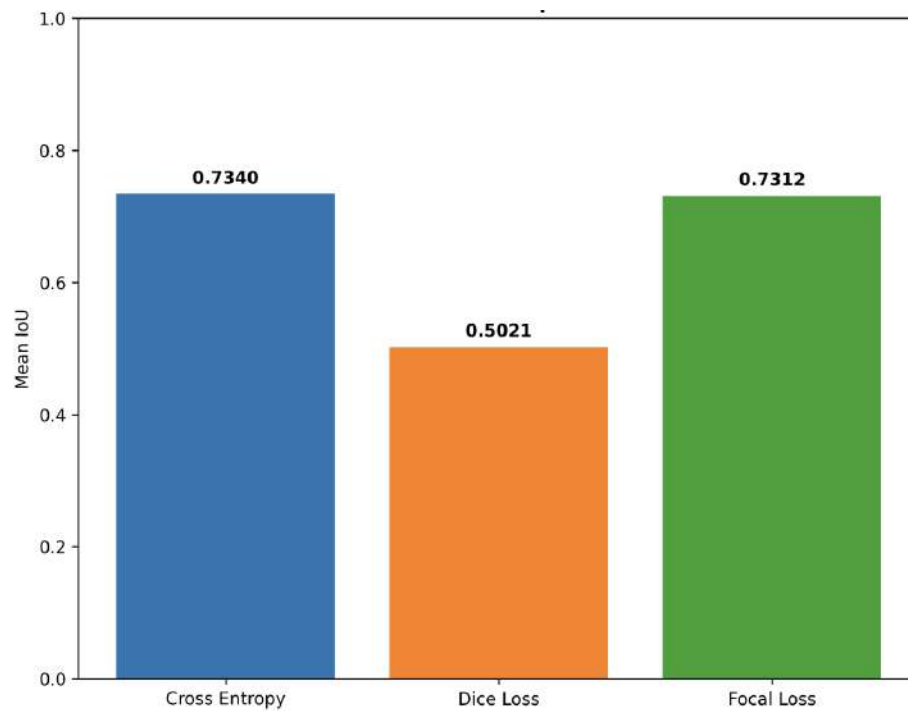


Figure A.6: Overview of OVSeg. Stage 1: a segmentation model produces mask proposals and embeddings with a mask loss ($Loss_{mask}$). Stage 2: CLIP is adapted using mask-category pairs, aligning masked crops with class prompts via a classification loss ($Loss_{cls}$), enabling open-vocabulary segmentation. Reproduced from [26]

A.2 SegFormer



(a) Validation IoU curves for SegFormer across training epochs using cross-entropy, dice, and focal loss.



(b) Final test IoU results for SegFormer under the three loss functions.

Figure A.7: Comparison of SegFormer segmentation performance with different loss functions. Subfigure (a) shows validation IoU trends during training, while subfigure (b) summarizes the final test IoU values.

Table A.2: Perturbations applied in robustness testing and their rationale.

Perturbation	Severity Levels	Rationale
Gaussian Noise	Variance: 0 to 18	Simulates sensor noise and low-light capture artifacts.
Gaussian Blur	Kernel size factor: 0 to 9	Models out-of-focus shots or camera shake.
Contrast Increase	Scale: 1.0 to 1.25	Emulates boosted contrast from harsh lighting.
Contrast Decrease	Scale: 1.0 to 0.10	Simulates low-contrast conditions from haze/fog.
Brightness Increase	Offset: 0 to +45	Models overexposure such as strong sunlight.
Brightness Decrease	Offset: 0 to -45	Simulates underexposure in low light.
Occlusion	Area occluded: 0% to 45%	Represents partial object coverage.
Salt & Pepper Noise	Noise fraction: 0.00 to 0.18	Emulates transmission errors and dropouts.

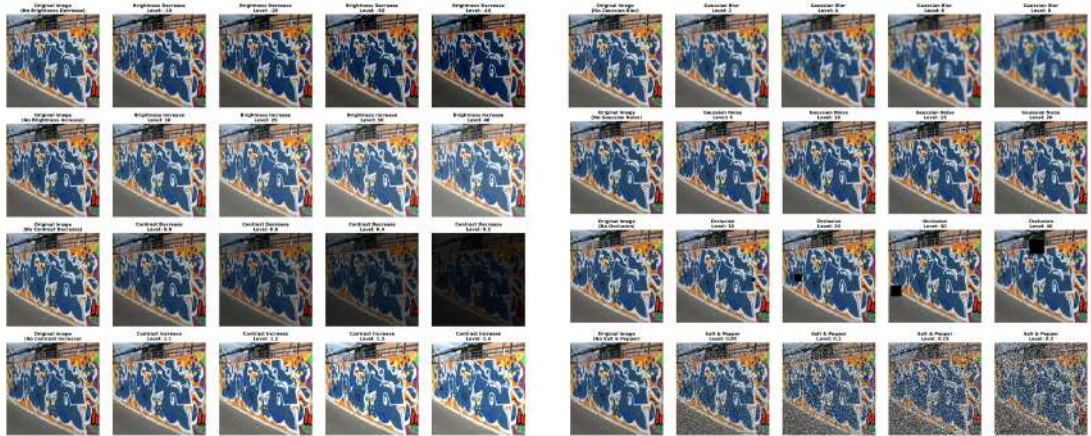


Figure A.8: Perturbation examples for robustness testing.

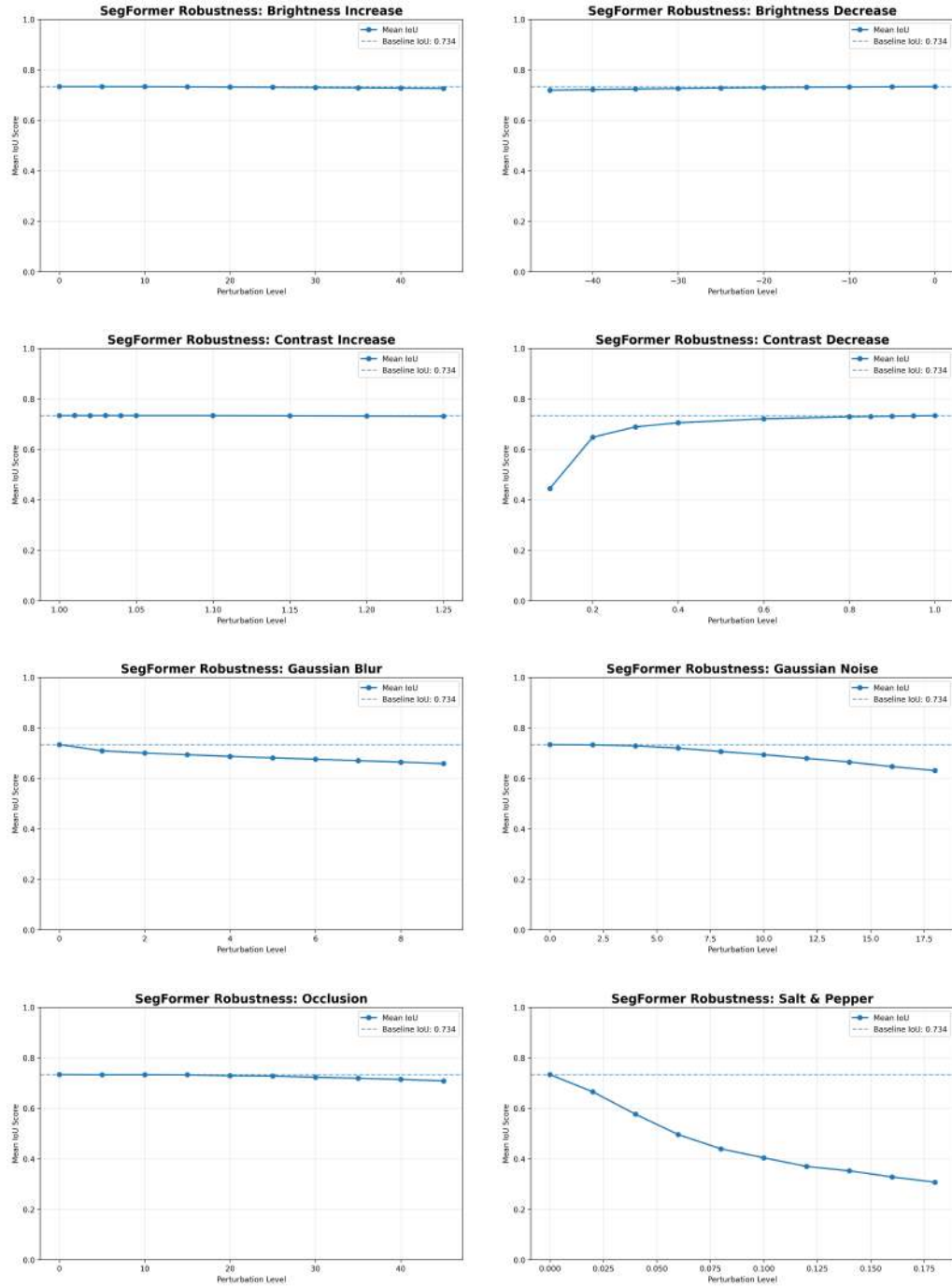


Figure A.9: Results of robustness testing under eight types of perturbations. Each plot shows the effect of increasing severity on segmentation performance.

A.3 CLIPSeg

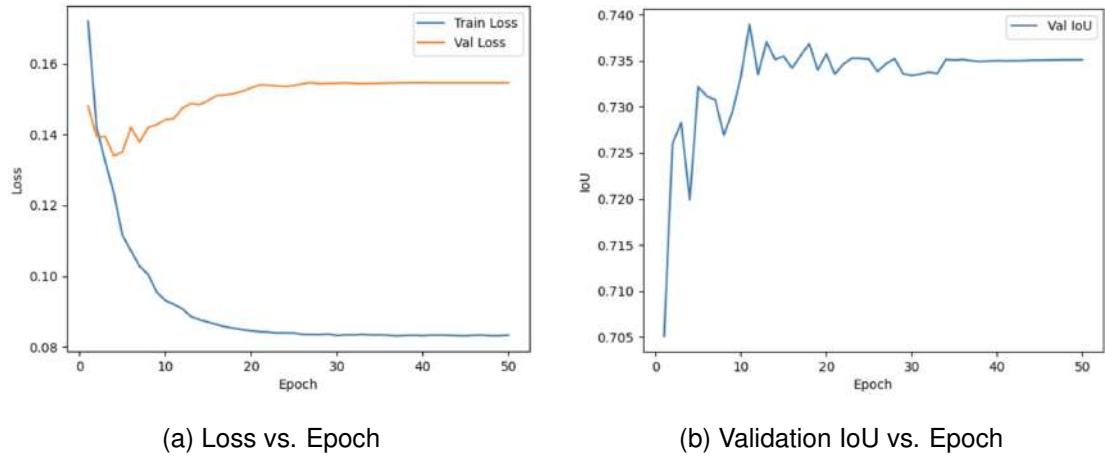


Figure A.10: Training results for CLIPSeg fine-tuning in the binary setting: (a) training & validation loss and (b) validation IoU over epochs.

Table A.3: Per-class mIoU before and after fine-tuning CLIPSeg (Binary setting). Improvement is absolute change in mIoU.

Class	Before	After	Improvement
Background	0.8028	0.8989	+0.0961
Pollutant	0.4802	0.7070	+0.2268

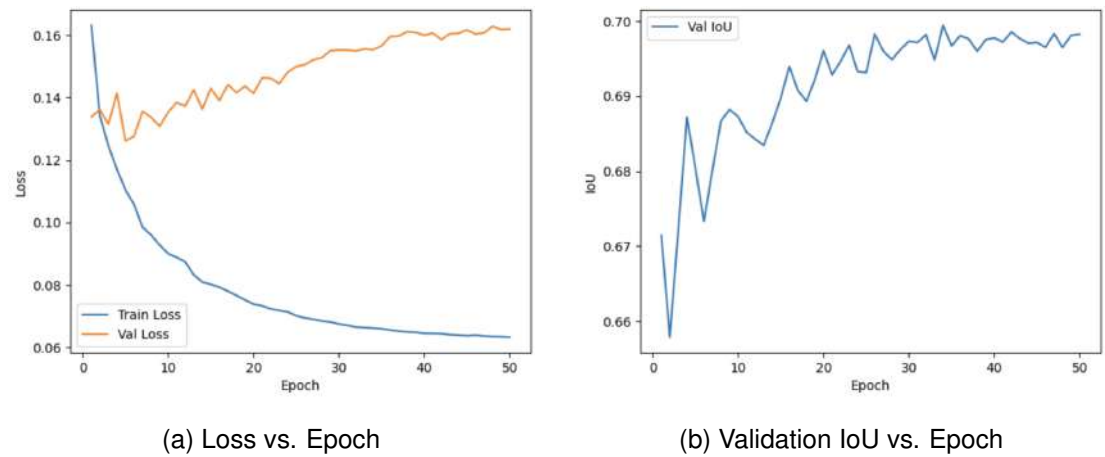
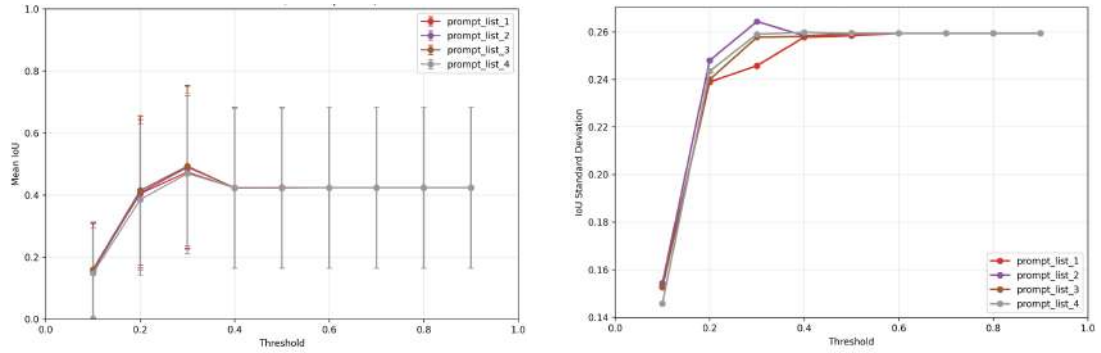


Figure A.11: Training results for CLIPSeg fine-tuning in the multi-class setting: (a) training & validation loss and (b) validation IoU over epochs.

Table A.4: Per-class mIoU before and after fine-tuning CLIPSeg (Multi-class setting). Improvement is absolute change in mIoU.

Class	Before	After	Improvement
Background	0.7880	0.8781	+0.0902
Billboard	0.4620	0.6906	+0.2286
Construction material	0.2706	0.7806	+0.5100
Garbage	0.6270	0.7324	+0.1054
Graffiti	0.7022	0.7870	+0.0849
Pothole	0.0299	0.4162	+0.3863
Wires	0.2656	0.3990	+0.1334

A.4 OVSeg



(a) Mean IoU as a function of probability threshold for different prompts.

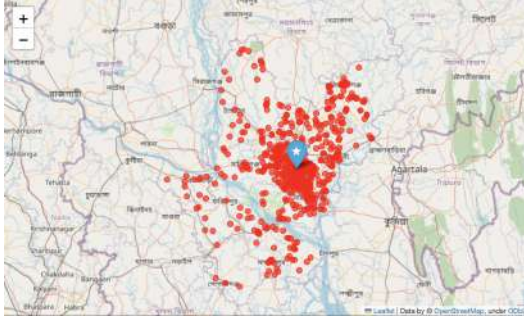
(b) IoU standard deviation as a function of probability threshold.

Figure A.12: Multiclass segmentation performance of OVSeg for four different prompts across nine probability thresholds.

Table A.5: Per-class IoU performance of OVSeg using threshold 0.3 and `prompt_list_3` (simple, single-token prompts).

Class	Mean IoU	Std IoU
Background	0.5395	0.2877
Billboard	0.5238	0.3208
Construction material	0.4483	0.3708
Garbage	0.4878	0.2817
Graffiti	0.5423	0.2946
Pothole	0.1070	0.1472
Wires	0.1906	0.1763

A.5 Visual Pollution Scoring



(a) Dhaka, Bangladesh



(b) Jakarta, Indonesia



(c) New York, USA



(d) Barcelona, Spain

Figure A.13: Randomly sampled coordinates within four cities, extracted from OpenStreetMap. These served as the basis for Street View image collection.



(a) Sample 1 — SWS = 0.68, CCR = 0.50, PR = 0.60, VPI = 0.38



(b) Sample 2 — SWS = 0.63, CCR = 0.67, PR = 0.70, VPI = 0.45

Figure A.14: Two example sets of randomly grouped images from Jakarta with corresponding segmentation predictions and VPI results.



(a) Sample 1 — SWS = 0.60, CCR = 0.33, PR = 0.70, VPI = 0.36

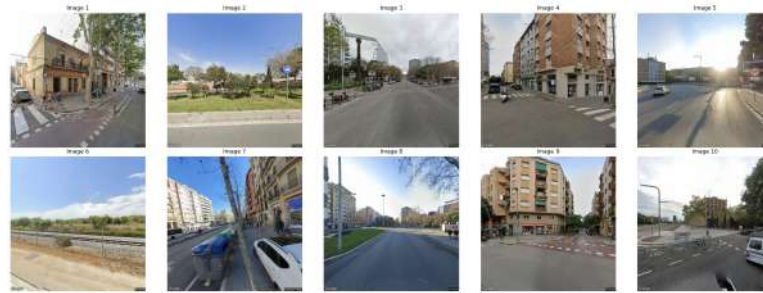


(b) Sample 2 — SWS = 0.60, CCR = 0.17, PR = 0.50, VPI = 0.24

Figure A.15: Two example sets of randomly grouped images from New York with corresponding segmentation predictions and VPI results.



(a) Sample 1 — SWS = 0.99, CCR = 0.33, PR = 0.20, VPI = 0.16



(b) Sample 2 — SWS = 0.80, CCR = 0.17, PR = 0.10, VPI = 0.06

Figure A.16: Two example sets of randomly grouped images from Barcelona with corresponding segmentation predictions and VPI results.